

Topological image feature extraction for Vision Transformer Breast Cancer Classification



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May 23, 2026



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Final Work Card

Title of the work:	Topological image feature extraction for Vision Transformer Breast Cancer Classification
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Date of delivery:	May 23, 2026
Studies or Program:	Master in Data Science
Area or the Final Work:	Machine learning in clinical data analytics
Language of the work:	English
Keywords:	Image classification, Deep learning, Breast cancer

Abstract

This project aims to develop a Deep Learning pipeline for the binary classification of mammographic images (tumor vs. non-tumor) using the CBIS-DDSM dataset. The proposed approach combines image preprocessing techniques, including noise reduction and contrast enhancement, with feature extraction based on Topological Data Analysis (TDA) to capture structural properties of breast tissue.

Both the processed images and the extracted topological descriptors will be integrated with a Vision Transformer (ViT) model for final classification. The objective is to evaluate whether incorporating explicit topological information improves classification performance.

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Chapter 1

Introduction

Breast cancer remains the most prevalent oncological disease worldwide. According to the World Health Organization [1], it represents a leading cause of cancer-related mortality among women globally. Early detection plays a crucial role in improving survival rates, and medical imaging techniques such as mammography are fundamental tools in preliminary diagnosis.

In recent years, significant advances in Artificial Intelligence and Deep Learning have transformed medical image analysis. New approaches such as lightweight architectures [2], hybrid feature extraction techniques [3], and transfer learning strategies [4] have demonstrated promising results in improving diagnostic performance while reducing computational cost. These developments open new possibilities for enhancing automated breast cancer detection systems.

Motivated by these advances and supported by a strong mathematical and analytical background, this work explores the integration of modern deep learning architectures with advanced image processing and structural feature extraction techniques for mammographic image classification.

1.1 Context and justification of work

Breast cancer diagnosis has become an active research area due to its medical relevance and the rapid development of computational methods for medical imaging analysis. The current research landscape shows a significant growth in the application of deep learning techniques for tumor detection and classification, including generative adversarial networks (GANs), convolutional neural networks (CNNs) and large language models (LLMs) [5].

Breast imaging can be obtained through several procedures: mammography, ultrasound, magnetic resonance imaging (MRI), synthetic phantoms, etc. Among publicly available datasets, two widely used resources are the Curated Breast Imaging Subset of Digital Database for Screening Mammography (CBIS-DDSM) [6] and the VICTRE breast phantom dataset [7]. Since mammography remains the most common imaging modality for preliminary breast cancer screening, this work focuses on the CBIS-DDSM dataset to ensure clinical relevance and comparability with previous studies.

Image preprocessing plays a critical role in image classification. Common techniques include noise reduction and contrast enhancement, which improve the visibility of relevant anatomical structures and lesions. Previous studies have demonstrated the effectiveness of such preprocessing steps in improving classification performance [8, 9]. Therefore, similar preprocessing

strategies are incorporated into the proposed pipeline.

In addition to conventional preprocessing, this work introduces Topological Data Analysis (TDA) as a complementary feature extraction approach. Recent research has shown promising results when integrating topological information into deep learning models [10]. TDA captures structural and connectivity-related properties of images that may not be fully represented by pixel-based learning alone. Furthermore, the use of explicit structural descriptors may contribute to improved explainability, which is particularly important in medical applications.

Regarding model selection, comparative studies evaluating different deep learning architectures for breast cancer classification indicate that Vision Transformers (ViT) achieve competitive or superior performance compared to traditional convolutional networks [11, 12]. Based on these findings, the Vision Transformer architecture is adopted as the primary classification model in this work.

1.2 Work objectives

The central hypothesis of this work is that the integration of image preprocessing techniques and Topological Data Analysis (TDA) features into a Vision Transformer (ViT) architecture improves the classification performance of mammographic images compared to a baseline ViT model without such enhancements.

More specifically:

- H1: Image preprocessing techniques (noise reduction and contrast enhancement) positively impact the classification accuracy of a Vision Transformer model.
- H2: The incorporation of topological descriptors derived from TDA provides complementary structural information that improves classification performance compared to a standard Vision Transformer.

To test these hypotheses, the following objectives are defined:

- To implement a baseline Vision Transformer model for mammographic image classification using the CBIS-DDSM dataset.
- To evaluate the impact of image preprocessing techniques on the performance of the baseline model.
- To extract topological features from mammographic images using Topological Data Analysis methods.
- To design the architecture integrating TDA features with the Vision Transformer model.
- To compare all approaches using quantitative performance metrics such as accuracy, sensitivity, specificity, F1-score, and AUC.
- To develop a theoretical and practical understanding of Vision Transformer architectures and medical image preprocessing techniques.

1.3 Impact on sustainability, ethics-social and diversity

This project is aligned with the United Nations Sustainable Development Goals (SDGs) [13], particularly Goal 3 (Good Health and Well-being) and Goal 10 (Reduced Inequalities).

Breast cancer is one of the leading causes of cancer-related mortality worldwide [1]. Early diagnosis and treatment through screening programs significantly increase survival rates. The development of computer-aided diagnostic tools based on Artificial Intelligence can support medical professionals by assisting in the early detection of suspicious lesions, potentially reducing diagnostic delays and improving patient outcomes (SDG 3).

From a social perspective, automated diagnostic support systems may help reduce inequalities in access to specialized medical expertise (SDG 10). In regions with limited access to experienced radiologists, AI-based tools could provide preliminary assessments, contributing to more equitable healthcare delivery.

However, the ethical implications of AI in medical decision-making must be carefully considered. Automated systems should not replace medical professionals but rather support them. Issues such as model bias, generalization across populations, transparency, and interpretability are critical. In this work, the incorporation of Topological Data Analysis aims not only to improve performance but also to enhance structural interpretability of the learned features.

Regarding sustainability, the computational cost of training deep learning models is also considered. The use of efficient architectures and controlled experimentation seeks to balance model performance with responsible resource consumption.

1.4 Focus and method followed

This work follows an experimental research strategy within the field of applied machine learning. The objective is to design, implement, and evaluate a deep learning-based classification system for mammographic images.

The research adopts a quantitative approach, where performance metrics obtained from experiments are used to validate the proposed hypotheses. Since the study focuses on computational model evaluation using publicly available datasets, no qualitative data collection methods are involved.

From a development perspective, this project can be classified as the design and implementation of a new computational pipeline integrating existing technologies (Vision Transformers and Topological Data Analysis) into a novel hybrid architecture.

The methodological process consists of the following stages:

1. Literature review on breast cancer image classification, preprocessing techniques, transformer-based architectures, and TDA.
2. Dataset selection and preparation using the CBIS-DDSM dataset.
3. Implementation of a baseline Vision Transformer model.
4. Integration of preprocessing techniques (noise reduction and contrast enhancement).
5. Extraction of topological descriptors and design of a hybrid model architecture.

6. Experimental evaluation and comparison of different model configurations using quantitative metrics.
7. Statistical analysis and interpretation of results.

The project follows an iterative prototyping methodology. Models are developed incrementally, starting from a baseline implementation and progressively integrating preprocessing and topological features. Each iteration includes training, validation, performance evaluation, and refinement.

The tools used include programming environments for deep learning model development, numerical computation libraries, and image processing frameworks appropriate for medical image analysis. In addition, a requirement analysis is conducted to clearly define the system specifications and guide the development process.

1.4.1 Requirement Analysis

The project requires the definition of both functional and non-functional requirements to ensure a structured development process.

Functional requirements:

- Load and preprocess the CBIS-DDSM dataset.
- Implement a Vision Transformer (ViT) for binary classification (benign vs malignant).
- Extract and integrate Topological Data Analysis (TDA) features.
- Train, validate, and test the model.
- Evaluate performance using metrics such as accuracy, precision, recall, and F1-score.

Non-functional requirements:

- Reproducibility of experiments.
- Computational efficiency.
- Modularity of the implementation.
- Interpretability of results.

1.5 Work schedule

The work schedule for this project has been structured to align with the defined objectives and deliverables. A Gantt diagram 1.1 has been created to break down each deliverable into individual tasks, illustrating their planned start and end dates, dependencies, and expected duration. The schedule is periodically updated to reflect the current progress, with the latest update recorded as of May 23rd, 2026.

This structured plan ensures systematic progress toward the project goals, facilitates monitoring of milestones, and allows timely adjustments if necessary.

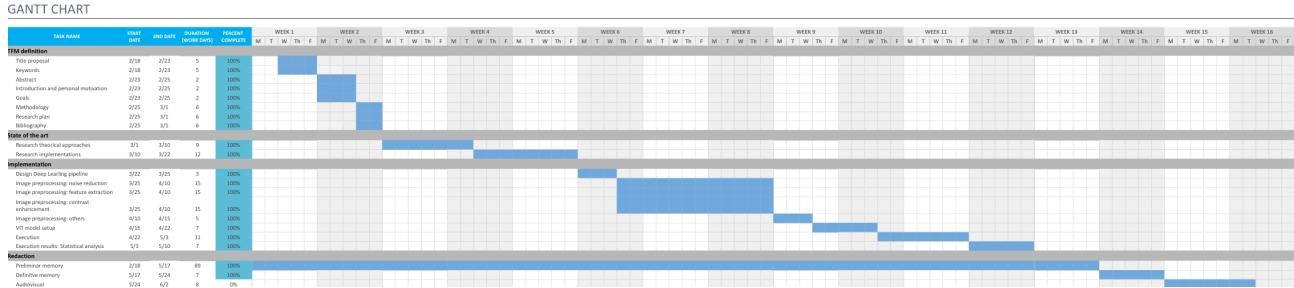


Figure 1.1: Gantt diagram

1.6 Brief summary of products obtained

By performing eight experiments, three different models were obtained: first, a Vision Transformer with different levels of preprocessing. Secondly, two Vision Transformers working in parallel to process two kinds of images and, lastly, a Vision Transformer combined with a linear model, that processes the topological descriptors of the images. The results are mixed: pre-processing consistently improves performance, while TDA features do not surpass the baseline. The best configuration achieves AUC comparable to existing literature.

1.7 Brief description of other memory chapters

Chapters are structured as follows:

1. **Introduction:** presents the project, hypothesis and goals.
2. **State of the Art:** covers the current status of the research and identifies the existing gaps.
3. **Materials and methods:** explains the experiments designed and implementation.
4. **Statement on the use of generative artificial intelligence:** to disclaim where and how GenAI has been used.
5. **Results:** answers the hypothesis defined in the first chapter.
6. **Discussion:** deeply analyzes the values obtained and their meaning.
7. **Conclusions and future works:** investigates what the next steps could look like.
8. **Glossary:** briefly defines the key terms and concepts used.

Chapter 2

State of the Art

In this chapter we will cover the State of the Art of each of the components involved in the model proposed in the Context and justification of work 1.1 above. This chapter is split into four sections:

1. **How CBIS-DDSM is used and evaluated in the literature:** describes the characteristics of the dataset and how the dataset has been studied. It also explores what techniques and metrics are used to evaluate the results and their meaning.
2. **Vision Transformers (ViT):** explains what Vision Transformers are, and compares them to classical Convolutional Neural Networks (CNNs).
3. **Topological Data Analysis (TDA) applied to images:** details what topological features are and how they can be extracted from images.
4. **Combining ViT and TDA:** explores the research gap and how this thesis can help reduce it.

Following these steps, it will provide a comprehensive review of the current status of the research related to the project.

2.1 CBIS-DDSM dataset

In this section we will explore how the CBIS-DDSM dataset is used and evaluated in the literature. First, let's recall the characteristics of the dataset: by visiting the documentation provided by the Cancer Imaging Archive [6], the DDSM is a database of 2,620 scanned film mammography studies. It contains normal, benign, and malignant cases with verified pathology information from 1566 subjects; and the details on how to use the dataset were published with the dataset [14].

The CBIS-DDSM dataset addresses a key limitation in previous research: the lack of standardized and reproducible evaluation protocols in mammography-based computer-aided diagnosis systems. Prior studies often relied on private datasets or non-standard subsets of public databases, making direct comparison between methods difficult. To overcome this issue, the

authors introduced a curated and standardized version of the original dataset: decompressed images in DICOM format, improved region-of-interest (ROI) annotations, and verified pathology labels. Additionally, predefined training and testing splits were provided to facilitate consistent evaluation across different studies.

Another important contribution of this work is the enhancement of annotation quality. The ROI segmentations were refined using automated methods and validated against expert radiologist annotations using metrics such as the Dice coefficient, showing a significant improvement over the original DDSM annotations. This increases the reliability of the dataset as a benchmark for both detection and segmentation tasks. Overall, CBIS-DDSM was designed not only as a larger and cleaner dataset, but as a step towards more reproducible and comparable research in breast cancer diagnosis using medical imaging.

This dataset has been used to study breast cancer from very different angles: from classification problems (binary [15] or multi-class [16]), localization [17], segmentation [18], etc.

In this paper, we will focus on binary classification; in the next paragraphs we visit different binary classification relevant papers that have used this dataset, what results have they obtained and what do these results mean. Before doing that, we cover some definitions regarding binary classification evaluation [15].

2.1.1 Evaluation Metrics

The metrics used in the papers that we have studied are the following: accuracy, F1 score and Area Under the Curve (AUC). However, other metrics like sensitivity and specificity are very meaningful in medical areas, as a false negative can have worse consequences than a false positive when diagnosing breast cancer images. All these metrics are covered in this section:

Confusion Matrix

A confusion matrix is a tabular representation that is used to analyse the performance of a classification model by comparing predicted labels with the true labels. In the case of binary classification, the confusion matrix consists of four main components:

- True Positive (TP): the number of instances that belong to the positive class and are correctly classified as positive.
- True Negative (TN): the number of instances that belong to the negative class and are correctly classified as negative.
- False Positive (FP): the number of instances that belong to the negative class but are incorrectly classified as positive.
- False Negative (FN): the number of instances that belong to the positive class but are incorrectly classified as negative.

Accuracy

Accuracy is one of the most commonly used evaluation metrics. It measures the proportion of correctly classified instances out of the total number of instances. It is defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (2.1)$$

Although accuracy is intuitive and easy to interpret, it may not be a reliable metric in the presence of class imbalance. In such situations, a model may achieve a high accuracy while still performing poorly on the minority class.

Sensitivity (Recall)

Sensitivity, also known as recall or True Positive Rate (TPR), measures the proportion of actual positive instances that are correctly identified by the model. It is defined as:

$$\text{Sensitivity} = \frac{TP}{TP + FN} \quad (2.2)$$

In the clinical context of breast cancer screening, sensitivity is particularly important because it reflects the model's ability to detect malignant cases. A low sensitivity implies a high rate of false negatives, meaning malignant lesions that are incorrectly classified as benign, which can result in delayed diagnosis and treatment.

Specificity

Specificity, also known as the True Negative Rate (TNR), measures the proportion of actual negative instances that are correctly identified by the model. It is defined as:

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (2.3)$$

Specificity reflects the model's ability to correctly identify benign cases: a low specificity means a high rate of false positives.

F1 Score

The F1 score is a metric that combines precision and recall into a single value, providing a balance between the two. It is particularly useful when both false positives and false negatives are of concern.

First, precision and recall are defined as follows:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2.4)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (2.5)$$

The F1 score is then computed as the harmonic mean of precision and recall:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (2.6)$$

This metric is especially appropriate in scenarios with imbalanced datasets, where a trade-off between precision and recall must be considered.

Area Under the Curve (AUC)

The Area Under the Curve (AUC) typically refers to the area under the Receiver Operating Characteristic (ROC) curve. The ROC curve illustrates the relationship between the True Positive Rate (TPR) and the False Positive Rate (FPR) across different classification thresholds.

The TPR and FPR are defined as:

$$\text{TPR} = \frac{TP}{TP + FN} \quad (2.7)$$

$$\text{FPR} = \frac{FP}{FP + TN} \quad (2.8)$$

The AUC value can be interpreted as the probability that the model assigns a higher score to a randomly selected positive instance than to a randomly selected negative instance. An AUC value close to 1 indicates strong discriminative ability, whereas a value close to 0.5 suggests performance equivalent to random classification.

2.1.2 Studies done with the CBIS-DDSM Dataset

Next, we compare some of the work that has been done with the dataset, to analyze how they processed the data, which models they trained and which results were achieved.

- **Transfer Learning and Fine Tuning in Breast Mammogram Abnormalities Classification on CBIS-DDSM Database** [15]: the methodology proposed in this paper prepares the data by changing the resolution of the images and augmenting the contrast using the Contrast Limited Adaptive Histogram Equalization (CLAHE) [19] that enhances image quality by applying histogram equalization locally on small regions (tiles), which improves contrast and reveals fine details in different parts of the image. Next, they find the region of interest (ROI) using a box surrounding the suspicious mass, and normalize that region again. Since these boxes can be of different shapes, they keep only the ROIs with an aspect ratio $r = \frac{w}{h}$ between 0.4 and 1.5. Lastly, the ROIs are resized to a final shape of 320×320 : they added a padding of 8 pixels and then cropped the center of the image, and filtered the image to reduce noise.

Next, a step of data augmentation is taken to prevent overfitting. This is achieved by applying rotations, shears, flips, zooms, etc. to the images. They considered the Train/Test split that the CBIS-DDSM dataset presents, but joined them to augment the data and then split them into different Train and Test sets once they had increased the size to 60 thousand images. They split it into 80% for training, 10% for validation and the last 10% for testing.

Lastly, they used the modified dataset in seven CNNs that were pretrained, and studied the performance of Transfer Learning. Since the problem approached in this paper is binary classification, they evaluated the results using a confusion matrix, Accuracy, F1 score and Area under the ROC curve. They saw best results in VGG16 [20], so they fine-tuned it adjusting the initial model and obtained values of $\text{AUC} = 0.844$, $F_1 = 0.85$, $\text{ACC} = 0.844$.

- **Sensitivity-Based Breast Cancer Characterization on CBIS-DDSM Mammography Dataset Using Fine-Tuned EfficientNet-B2 Model** [21]: the preprocessing in this paper is specially different to the one we have seen above. The images are reduced to a constant resolution and no data augmentation processes were applied. Once more, they did not respect the Train-Test partition provided by the CBIS-DDSM dataset, but defined their own that was more balanced.

Since the amount of images corresponding to benign tumors is greater than the ones depicting malignant tumors, they opted for imbalance-conscious learning methods: an EfficientNet-B2, that was pretrained, followed by Global Average Pooling, a Fully Connected layer, Batch Normalization and a Sigmoid output layer. They fine-tuned the last 25 layers of the model and obtained $AUC = 0.679$, $ACC = 0.6496$, which is not bad considering that it had a small amount of trainable parameters.

- **Deep BI-RADS Network for Improved Cancer Detection from Mammograms** [22]: the uniqueness of this paper, compared with the two above, is that they do not use the full CBIS-DDSM dataset, but instead they take a subset and enrich it with additional clinical data: it utilizes the textual metadata associated with each mammogram, that provides extra information about the kind of lesion and breast descriptors of the Breast Imaging Reporting and Data System (BI-RADS) [23].

The architecture consists of two branches, each composed of 6 stacked identical attention-based layers, that takes images and textual descriptors. They process the images by scaling them to 1024×1024 pixels, and conduct data augmentation by doing flips and elastic deformation. Then, features are extracted from the images using BiT layers, that are pre-trained. That is followed by various attention layers, that range from Cross-Attention, Self-Attention, etc. The data is fed to them in mini-batches of 16. The best results obtained in this paper, that tried various (Query, Key, Value) configurations is $AUC = 0.878$, $F_1 = 0.780$, $ACC = 0.796$.

The following table summarizes the lines above.

Table 2.1: Summary of CBIS-DDSM binary classification studies

Study	AUC	ACC	F_1
Transfer Learning and Fine Tuning in Breast Mammogram Abnormalities Classification on CBIS-DDSM Database [15]	0.844	0.844	0.85
Sensitivity-Based Breast Cancer Characterization on CBIS-DDSM Mammography Dataset Using Fine-Tuned EfficientNet-B2 Model [21]	0.679	0.6469	-
Deep BI-RADS Network for Improved Cancer Detection from Mammograms [22]	0.878	0.796	0.780

From all these papers we can learn that there is not one standardized way to treat the volume of data in the CBIS-DDSM dataset, nor the imbalance that it presents. All of them

use pre-trained models and fine-tune them in the last step in order to achieve the best results, which is aligned with the plan laid out in this project. It is notable how the projects that augmented the data and added other forms of data such as text obtained the best results, while the project that kept the dataset intact presented worse AUC and ACC.

2.2 Vision Transformers (ViT)

This section introduces Vision Transformers (ViT) and explains how they operate. They are also compared with traditional Convolutional Neural Networks (CNNs), which are the more traditional alternative to address image classification problems like the one proposed in this project.

Vision Transformers were introduced in 2021 in the paper *An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale* [24]. The paper studies how Transformer architectures, which are very successful in Natural Language Processing, can be adapted to Computer Vision tasks.

The idea proposed by Dosovitskiy et al. [24] is to remove convolution completely and instead apply the transformers directly to images, that are codified as sequences of tokens. This new approach reduces the resources needed, whilst maintaining excellent results when pre-trained with large amounts of data. Consequently, we can split Vision Transformers into two stages:

1. **Image to patch embeddings:** to create a one-dimensional sequence of token embeddings to feed the transformer from a 2D image, ViT first converts an image into a sequence of patches. Given an input image $x \in \mathbb{R}^{H \times W \times C}$, where (H, W) is the resolution of the image and C is the number of channels (3 in case of an RGB image), the image is divided into $N = \frac{HW}{P^2}$ 2D non-overlapping patches of size $P \times P$: $X_p \in \mathbb{R}^{N \times P^2 \times C}$. N will be the input sequence length for the Transformer.

Each of the transformer layers uses a constant latent vector of size D : the N image patches need to be flattened into D dimensions using a trainable linear projection, that results into the *patch embedding*, similar to word embeddings in NLP.

$$z_0 = [x_{\text{class}}; x_p^1 E; x_p^2 E; \dots; x_p^N E] + E_{\text{pos}}$$

where $E \in \mathbb{R}^{(P^2 \cdot C) \times D}$, $E_{\text{pos}} \in \mathbb{R}^{(N+1) \times D}$ and x_{class} is a learnable classification token.

Position embeddings are added to the patch embeddings to retain positional information. It is done using standard learnable 1D position embeddings. The resulting sequence of embedding vectors serves as input to the encoder.

2. **Transformer encoder:** after embedding, the sequence of patches is passed to a standard Transformer encoder. The encoder consists of multiple identical layers, each containing:
 - Multi-Head Self-Attention (MSA): introduced in *Attention Is All You Need* [25], it is a mechanism to compute attention between all input tokens using multiple parallel attention heads to capture diverse relationships.
 - Feed-Forward Network (MLP): consisting of fully connected layers with GELU activation functions [26]: it consists in non-linear Gaussian activation.

- Layer Normalization (LN) before every block.
- Residual connections after every block.

The work done in *An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale* [24] found that the Vision Transformers are, in general, less robust to small perturbations than conventional CNNs: they have less inductive bias. This is because, at the initialization time, the position embeddings carry no information about the positions of the patches. Since they have to be learned from scratch, and only MLP layers are invariant to translations. Dividing the image into patches breaks the two-dimensional neighborhood structure but, when trained with large datasets, having learned these spatial relationships from the data makes it more flexible.

For these reasons, the typical use case of a ViT requires pre-training with a big dataset and then fine-tuning to the smaller task (breast cancer image classification). The initial study [24] found that pre-trained ViTs dominate CNNs on the performance/compute trade-off.

This original proposal for Vision Transformers [24] compared them to Residual Networks (ResNets), those are a particular type of CNNs introduced in 2015 by Kaiming He et al., in a paper titled *Deep Residual Learning for Image Recognition* [27]. ResNets are designed to enable the effective training of very deep models by introducing *residual learning*. Instead of directly learning a desired mapping $H(x)$, each block learns a residual function $F(x) = H(x) - x$, which is combined with the input through identity shortcut (skip) connections. The results show that gradients flow more easily through the network; and it mitigates the degradation problem where deeper networks perform worse than shallower ones.

As it is a type of Convolutional Neural Network, it processes images by using convolutional layers with small kernels (e.g., 3×3). The main trait of them, as we just mentioned, is that the input is added back in every convolutional block. On the other hand, Vision Transformers do not use convolution whatsoever, and trade off that local context for global context. The following figure 2.1 illustrates these differences.

2.3 Topological Data Analysis (TDA) applied to images

This section explains what topological descriptors represent, and how to compute homology descriptors that can be embedded in a Vision Transformer: persistence images.

Topology is a branch of mathematics that studies the shape of objects and their main properties that do not change under continuous transformations. In the context of topological data analysis (TDA), these concepts can be applied to study the structure of grayscale images, where the pixel intensity values define the data representation. One of the most commonly used techniques in TDA is persistent homology, which allows one to track the appearance and disappearance of topological features, such as connected components and holes, across different intensity thresholds.[28]

To obtain topological descriptors from a grayscale image, we consider each pixel to be associated with an intensity value. This image can be interpreted as a function defined on a grid, in which the intensity values represent different levels of the structure. To analyze this image using TDA, we study how the structure changes when we progressively vary a threshold over the intensity values.

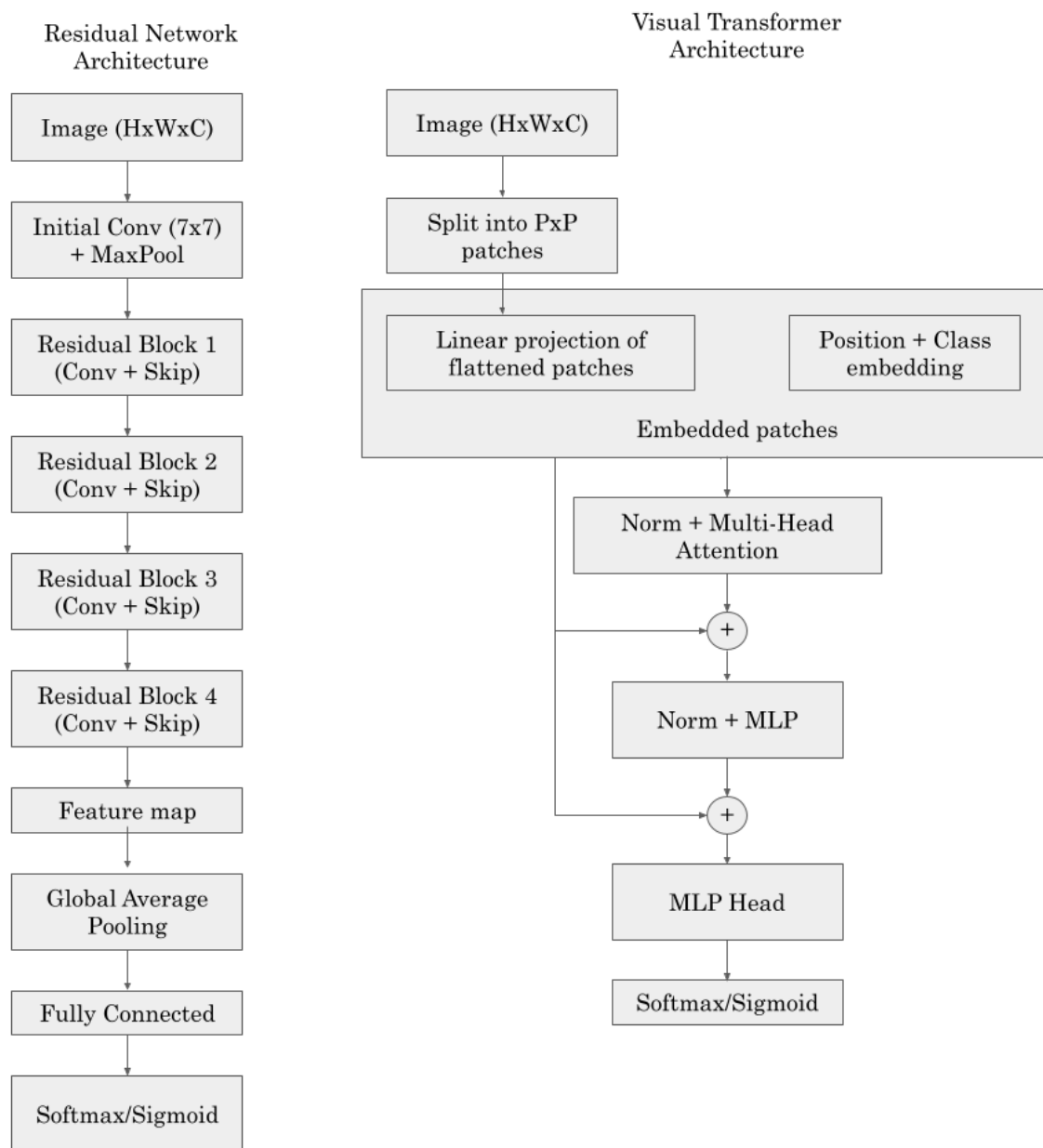


Figure 2.1: Residual Network and Vision Transformer Architectures

For each threshold value, we construct a binary image by selecting the pixels that satisfy the given condition. In this way, we can observe how many connected components and holes are present at each stage. As the threshold changes, new components can appear, existing ones can merge, and holes can be created or filled. Persistent homology allows to track these changes by recording the birth and death of each topological feature.

These features are represented using persistence diagrams (PDs), which describe the lifetime of each topological feature from its birth to its death: each point in the diagram with coordinates (b_i, d_i) corresponds to a feature with birth b_i and death d_i . These diagrams provide a compact summary of the topological structure of the image across all thresholds. Since PDs are not directly suitable for most machine learning models, they are usually transformed into finite-dimensional representations. This project uses the following:

1. **Betti curves:** show how many descriptors are alive at any given time. They can be computed for the topological descriptors of each homology class: in grayscale images: H_0 and H_1 . These curves can be discretized and fed as vectors to classifier models.[28]
2. **Landscapes:** given a persistence diagram, the landscapes can be computed by rotating it and laying it against the diagonal. Then, tent functions are computed for every point above the diagonal: lines parallel to the original axis that connect the point with the diagonal. The landscape functions can be discretized and fed as vectors to classifier models.[28]
3. **Silhouettes:** are a weighted average of tent functions with weights $w_i > 0$. A popular choice is $w_i = (d_i - b_i)^p$ where (b_i, d_i) is each point of the persistence diagram and p is a positive integer. Again, these functions can be discretized and fed as vectors to classifier models.[28]
4. **Persistence images (PIs):** the persistence diagram is converted into a persistence image by mapping each point into a continuous domain (usually transposing the persistence diagram) and applying a suitable probability distribution and weighting function, $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, that must be zero along the horizontal axis, continuous and piecewise differentiable[29]. The example in [29, Image 1] illustrates this procedure.

Figure 2.2 illustrates this process with an example grayscale image.

2.4 Combining ViT and TDA

Finally, this section explores existing systems that use Topological Data Analysis to improve a Vision Transformer, how it is embedded into the architecture, what applications were explored, and the results achieved.

Recent works have shown that, while Vision Transformers are powerful in capturing global dependencies through self-attention, they may lack robustness in representing intrinsic geometric and topological structures of the data. The integration of TDA can help address this limitation, providing complementary information about the shape and connectivity of data distributions.

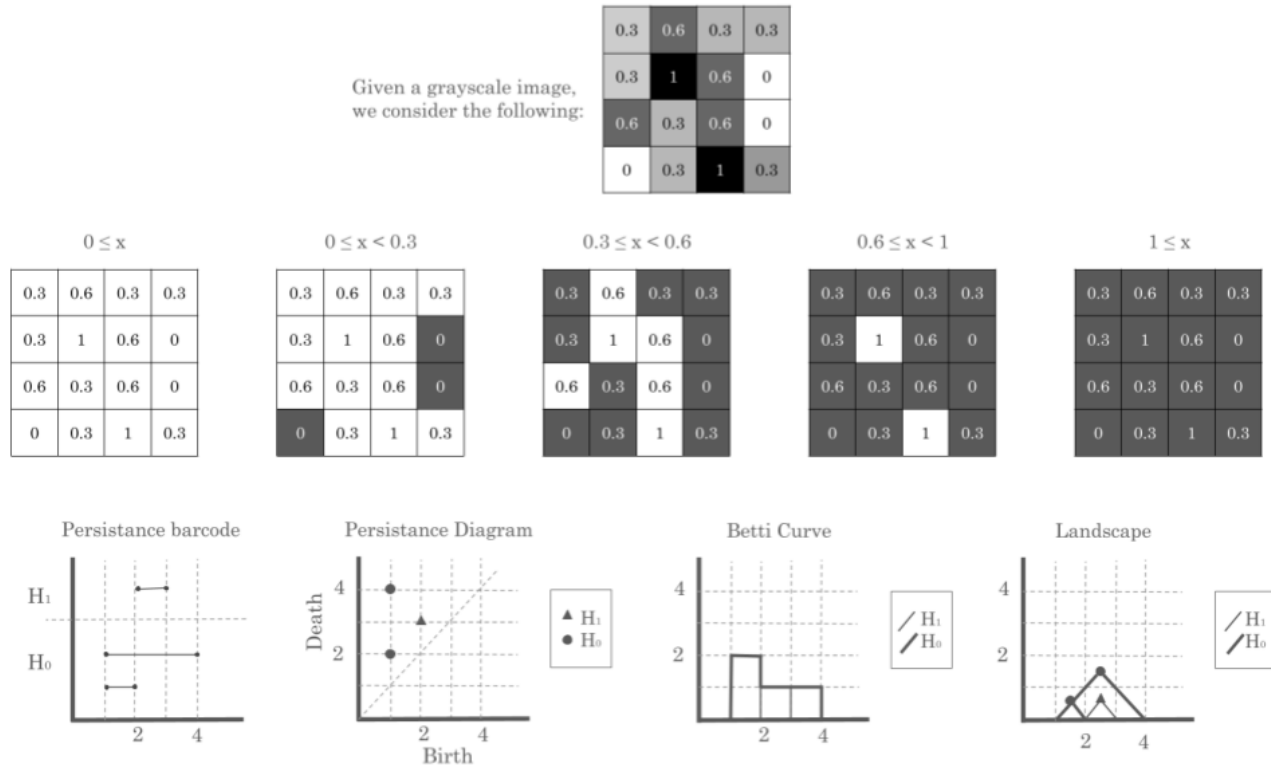


Figure 2.2: How to compute a persistence diagram: given a grayscale image, we consider how many connected components (H_0) and cycles (H_1) there are. By monitoring when they appear (birth) and when they get absorbed by another component, we can draw a Barcode Diagram, that has one parallel line for each component, and a Persistence Diagram, that has one point in coordinates (B, D) corresponding to each birth (B) and Death (D) of a component.

The work in *TDA-ViT: A Transformer-Based Framework for Unified Urdu Text Recognition via Topological and Visual Feature Fusion* [30] introduces a Transformer-based framework that combines visual and topological features for the task of Urdu text recognition. The approach followed consists of four stages:

1. Topological descriptors are extracted from images of Urdu text, and stored into persistence images 2.3.
2. Raw and persistent images are encoded using a Vision Transformer for feature embedding. This produces a high-dimensional feature sequence that encodes both pixel-level and topological information.
3. The resulting sequences are passed to a T5-based encoder (Transformer used for Natural Language Processing).
4. Lastly, the sequences are decoded using a GPT-2 based decoder.

This hybrid approach improves robustness to noise and distortions, which are common in handwritten or low-quality text images. Experimentally, the model demonstrates improved recognition accuracy compared to standard ViT-based baselines, highlighting the benefit of incorporating topological descriptors.

Despite these promising contributions, there are still several limitations and open challenges that motivate further research. First, the application domains explored in the current literature remain relatively limited, with a strong focus on text recognition. There is a lack of investigation in other domains such as medical imaging, where topological structures are often highly informative. In particular, mammography datasets such as CBIS-DDSM contain subtle structural patterns (e.g., mass irregularity, spiculation, and boundary complexity) that are not explicitly addressed in the existing works.

Second, other topological descriptors coming from images can be combined with Vision Transformers via vector composition, and that has not been broadly addressed in the papers found.

Lastly, the computational cost and scalability of TDA components are not deeply analyzed in these works. Persistent homology can be expensive to compute, especially for high-resolution medical images, and its impact on training efficiency remains an open question.

There is a gap in the literature regarding the design of TDA-enhanced Vision Transformer architectures for medical image classification. In this project, we aim to address it by investigating how topological features can be integrated into a ViT-based pipeline for binary classification of mammography images from the CBIS-DDSM dataset into benign and malignant categories.

Chapter 3

Materials and methods

This chapter describes the design and implementation of the experiments developed to evaluate the two hypotheses stated in Section 1.1. It covers the data preparation process, the image preprocessing techniques applied, the extraction of topological descriptors, the model architectures designed, and the experimental protocol followed. All experiments were implemented in Python using PyTorch [31] as the deep learning framework.

3.1 Data preparation

This section covers in detail how all data is obtained and prepared before starting the preprocessing.

3.1.1 Dataset selection and scope

As introduced in Section 2.1, this project uses the CBIS-DDSM dataset [6] for binary classification of mammographic lesions into benign and malignant categories. The dataset provides both breast masses and breast calcifications; however, this work focuses exclusively on the **breast mass** subset to maintain a controlled experimental scope and reduce computational cost.

The CBIS-DDSM dataset provides, for each lesion, three types of images: the full mammogram, a pre-cropped region of interest (ROI), and a binary ROI mask. The pathology labels include **MALIGNANT**, **BENIGN**, and **BENIGN_WITHOUT_CALLBACK**. Following the simplification adopted in the literature [15], the latter two categories are merged into a single **BENIGN** class, resulting in a binary classification task.

3.1.2 Data loading and file resolution

The CBIS-DDSM images are stored in DICOM format. Each DICOM file is loaded using the `pydicom` library [32], and the pixel array is normalized to the $[0, 255]$ range as an unsigned 8-bit integer:

$$I_{\text{norm}} = \left\lfloor \frac{I - I_{\min}}{I_{\max} - I_{\min}} \cdot 255 \right\rfloor \quad (3.1)$$

During the exploratory analysis, two issues were identified in the dataset file structure:

1. The file names provided in the CSV metadata do not match the actual file names on disk (the CSV references 000000.dcm and 000001.dcm, but the actual files have UUID-based names).
2. The ROI mask and the cropped image often share the same folder, making them indistinguishable by path alone.

To resolve these issues, the implementation identifies files by their **size**: within a shared folder, the smaller file corresponds to the cropped mammogram (fewer pixels), while the larger file corresponds to the ROI mask (which covers the full mammogram dimensions). This heuristic was validated manually and proved reliable across the entire dataset.

3.1.3 Train, validation, and test splits

The CBIS-DDSM dataset provides predefined training and testing partitions split by patient, ensuring that no patient appears in both sets. This project respects these original partitions to maintain comparability with the literature.

From the training partition, a stratified 85/15 split is performed to create a validation subset, using `StratifiedShuffleSplit` from `scikit-learn` [33] with a fixed random seed of 2 for reproducibility. The stratification ensures that the class distribution (benign vs. malignant) is preserved in both the training and validation subsets.

The resulting approximate sizes for the breast mass subset are:

- Training set: ~587 samples (85% of the original training partition).
- Validation set: ~104 samples (15% of the original training partition).
- Test set: ~201 samples (the original test partition).

3.1.4 Image caching

To avoid repeated DICOM parsing and file resolution during training, all cropped mammogram images are precomputed and stored as NumPy `.npy` arrays. A manifest CSV file (`cached.csv`) records the path and label for each sample, enabling efficient data loading during experimentation.

3.2 Image preprocessing

The preprocessing pipeline is designed to enhance the features in mammographic images and to reduce noise. It consists of two sequential operations applied to the grayscale image, and follows what was done in the other papers explored:

1. **Contrast Limited Adaptive Histogram Equalization (CLAHE)** [19]: is applied to enhance local contrast. Unlike standard histogram equalization, CLAHE operates on small tiles of the image and limits the amplification of contrast to avoid noise enhancement. The parameters used are: Clip limit: 2.0 and Tile grid size: 8×8 . These values are standard in the medical imaging literature and were adopted from previous work on the CBIS-DDSM dataset [15].

2. **Gaussian denoising:** after contrast enhancement, a Gaussian blur with a 5×5 kernel is applied to reduce high-frequency noise introduced by the CLAHE step. The standard deviation is computed automatically by OpenCV [34] based on the kernel size.

The complete preprocessing function applies CLAHE first, followed by Gaussian denoising. Figure 3.1 illustrates the effect of this pipeline on a sample mammogram.

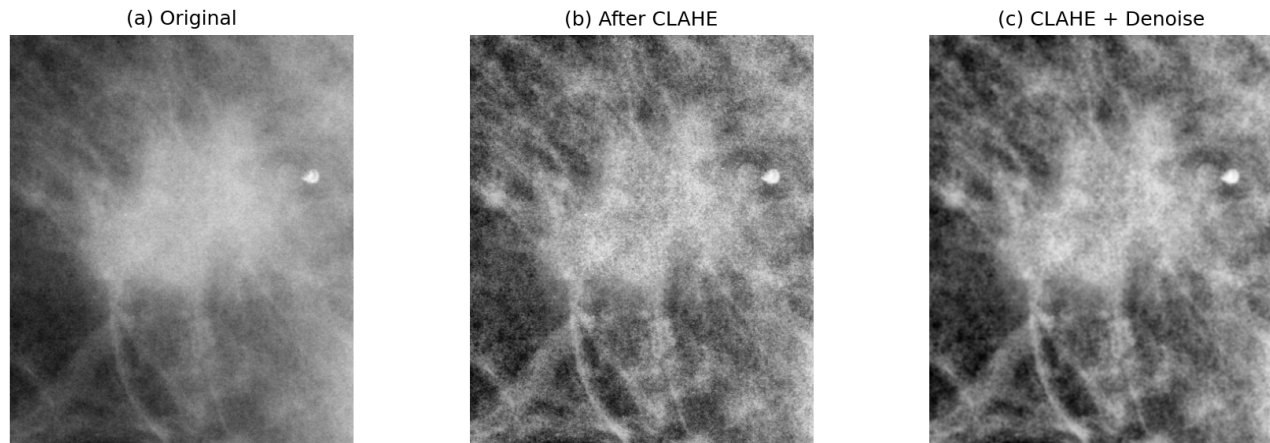


Figure 3.1: Effect of the preprocessing pipeline on a sample cropped mammogram: (a) original image, (b) after CLAHE, (c) after CLAHE + Gaussian denoise.

3.3 Topological feature extraction

This section describes how topological descriptors are extracted from mammographic images using the persistent homology framework introduced in Section 2.3. Two types of topological representations are computed: persistence images (for the dual-ViT architecture) and compact vector descriptors (for the linear fusion architecture). Details on those architectures are found in the next section 3.4.

Topological features are computed from two different image sources to evaluate which provides more informative structural information:

1. **Pre-cropped lesion:** the rectangular ROI crop provided by the dataset, which contains the lesion and some surrounding tissue.
2. **Masked mammogram crop:** the full mammogram multiplied by the binary ROI mask, then cropped to the bounding box of the mask with 15% padding. This isolates the lesion boundary more precisely.

3.3.1 Persistence images

Persistence images are computed using the `giotto-tda` library [35]. The process follows these steps:

1. **Cubical persistence:** the grayscale image is interpreted as a cubical complex, where pixel intensities define the filtration values. The `CubicalPersistence` class computes persistent homology for dimensions 0 (connected components) and 1 (cycles/holes).
2. **Persistence image transformation:** the resulting persistence diagram is transformed into a fixed-size persistence image of resolution 100×100 using the `PersistenceImage` class, which applies a Gaussian kernel and a linear weighting function as described in [29].

The output is a tensor of shape $(2, 100, 100)$: one 100×100 image for H_0 features and one for H_1 features. Figure 3.2 shows persistence images computed from a breast mass.

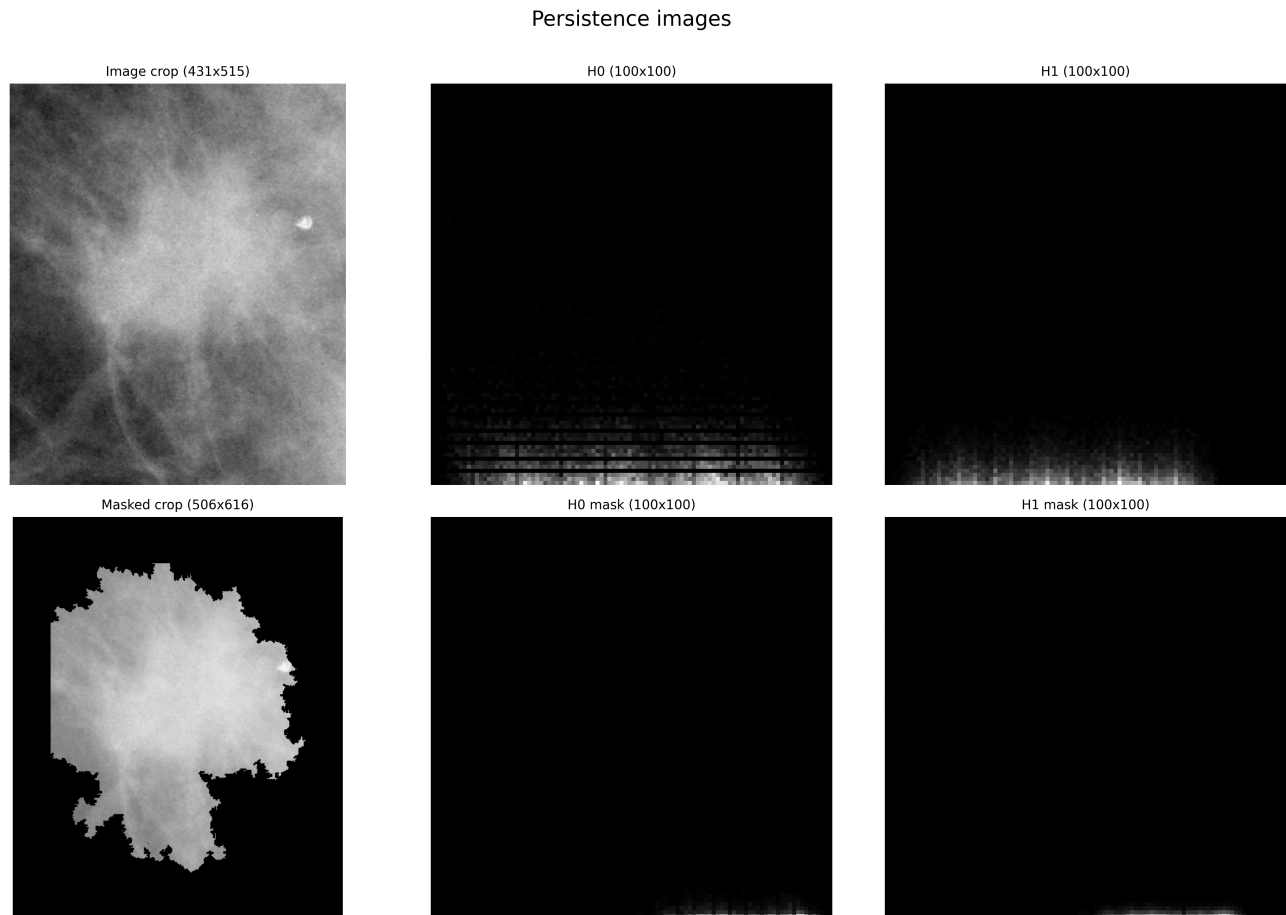


Figure 3.2: Persistence images computed on the cropped image (first row) and masked image (second row) of a breast mass.

3.3.2 Compact vector descriptors

Compact vector descriptors are computed using the `gudhi` library [36][37]. For each image, a cubical complex is constructed and persistent homology is computed for dimensions 0 and 1. Infinite persistence pairs (features that never die) are excluded. Three types of vectorizations are then computed at a resolution of 100 sample points:

- **Betti curves** [28]: discretized functions counting the number of features alive at each filtration value. One curve per homology dimension: $2 \times 100 = 200$ values.
- **Persistence landscapes** [28]: five landscape functions per homology dimension, each discretized at 100 points: $2 \times 5 \times 100 = 1000$ values.
- **Persistence silhouettes** [28]: weighted averages of tent functions, one per homology dimension: $2 \times 100 = 200$ values.

The three vectorizations are concatenated into a single descriptor of dimension $200 + 1000 + 200 = 1400$. Figure 3.3 shows persistence images computed from a breast mass, similar to the theoretical example given in figure 2.2.

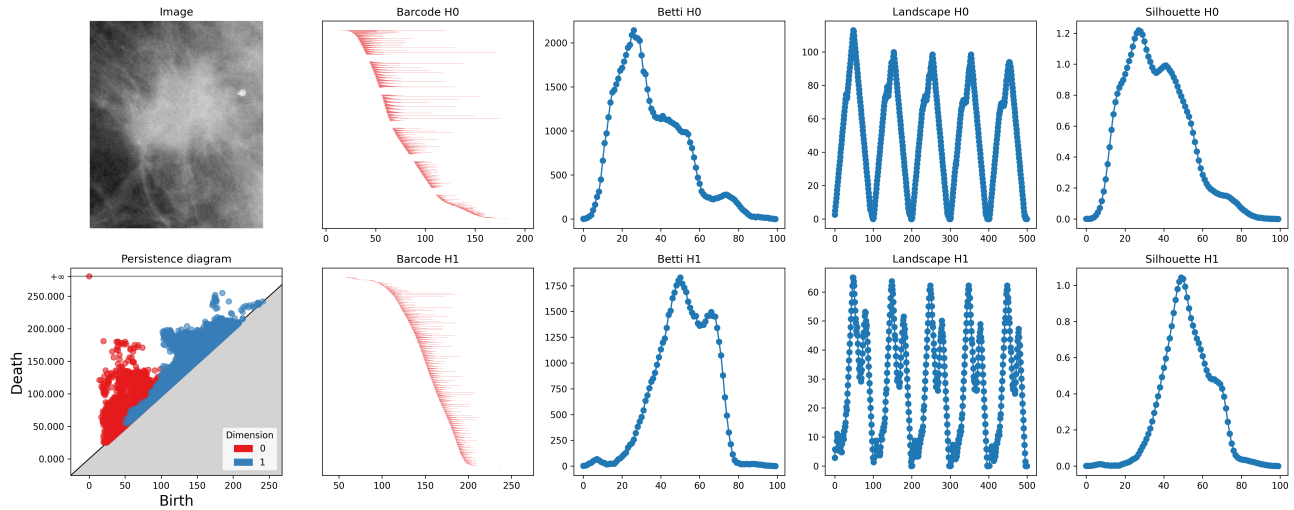


Figure 3.3: Persistence descriptors computed on the cropped image.

Since TDA computation is expensive (cubical persistence on high-resolution images), all topological descriptors are precomputed and cached as `.npy` files before training begins. The manifest CSV is extended with columns pointing to the corresponding TDA file for each sample.

3.4 Model architectures

Three model architectures are used across the eight experiments. All are based on the Vision Transformer (ViT-B/16) [24] pretrained on ImageNet, loaded via the `timm` library [38].

3.4.1 ViT classifier

The baseline architecture uses the standard ViT-B/16 model with its classification head replaced by a single linear layer that outputs one logit for binary classification:

$$\hat{y} = W \cdot z_{\text{CLS}} + b \quad (3.2)$$

where $z_{\text{CLS}} \in \mathbb{R}^{768}$ is the CLS token output from the transformer encoder, and $W \in \mathbb{R}^{1 \times 768}$, $b \in \mathbb{R}$ are the learnable parameters of the classification head.

Input images are resized to 224×224 pixels and replicated across three channels (since the original images are grayscale) to match the pretrained model's expected input format.

Figure 3.4 illustrates this architecture.

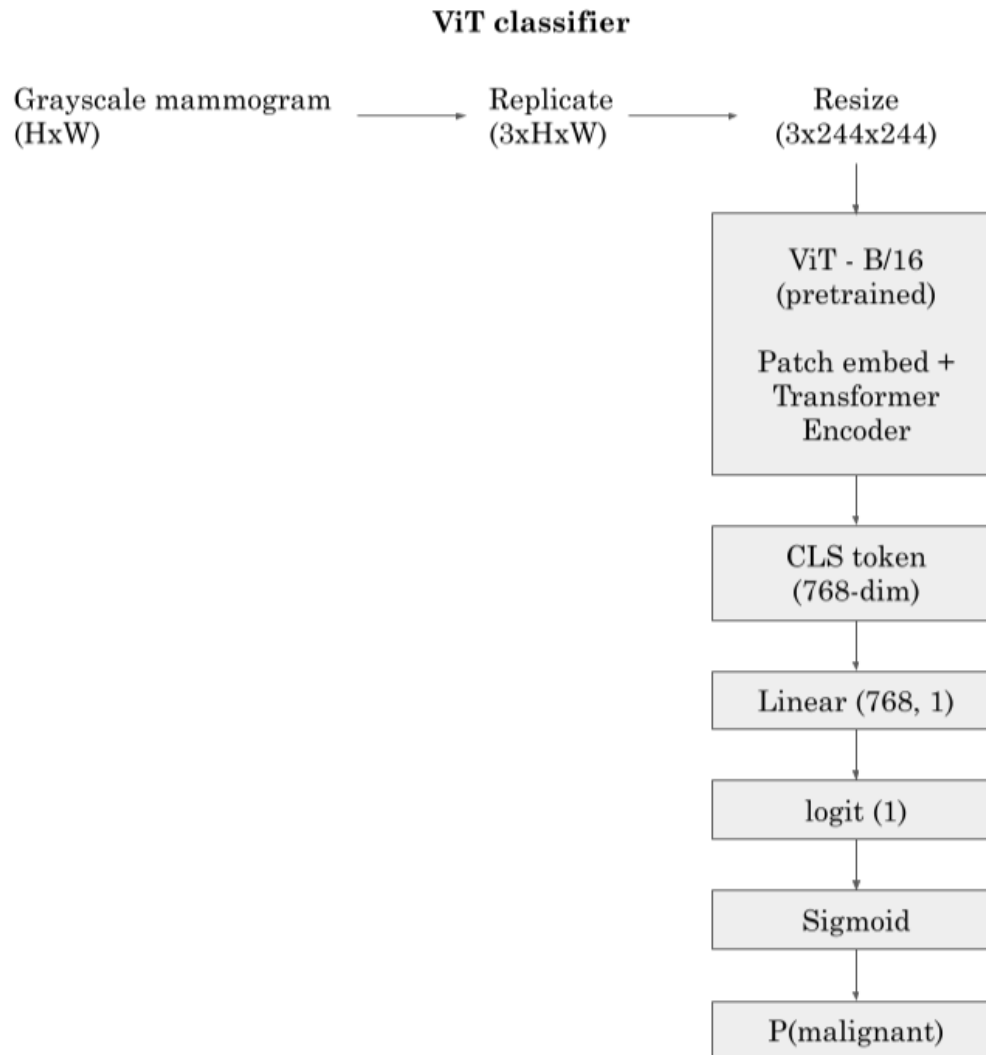


Figure 3.4: ViT classifier architecture: the grayscale mammogram is replicated to 3 channels, resized to 224×224 , and passed through the pretrained ViT-B/16. The CLS token is projected to a single logit via a linear layer.

3.4.2 Dual-ViT fusion model

This architecture processes the mammogram and its persistence image through two separate ViT-B/16 branches. The persistence image (shape $2 \times 100 \times 100$) is padded with a zero channel to form a 3-channel input and resized to 224×224 via bilinear interpolation. The CLS tokens from both branches are concatenated and passed through a classifier head:

$$\hat{y} = \text{MLP}([z_{\text{CLS}}^{\text{img}} \| z_{\text{CLS}}^{\text{TDA}}]) \quad (3.3)$$

where $\|$ denotes concatenation, and the MLP consists of:

$$\text{Linear}(1536, 256) \rightarrow \text{ReLU} \rightarrow \text{Linear}(256, 1)$$

Figure 3.5 illustrates this architecture.

3.4.3 Linear fusion model

This architecture combines the ViT image branch with a linear projection of the compact TDA vector descriptor. The ViT branch extracts the 768-dimensional CLS token from the mammogram. The TDA branch projects the 1400-dimensional vector through a linear layer to 128 dimensions. Both representations are concatenated and classified:

$$\hat{y} = \text{MLP}([z_{\text{CLS}}^{\text{img}} \| W_{\text{TDA}} \cdot v_{\text{TDA}}]) \quad (3.4)$$

where $v_{\text{TDA}} \in \mathbb{R}^{1400}$ is the compact descriptor, $W_{\text{TDA}} \in \mathbb{R}^{128 \times 1400}$ is the linear projection, and the MLP is:

$$\text{Linear}(768 + 128, 256) \rightarrow \text{ReLU} \rightarrow \text{Linear}(256, 1)$$

Figure 3.6 illustrates this architecture.

3.5 Data augmentation

Two levels of data augmentation are implemented, using transformations that are appropriate for mammographic images, which have no inherent orientation preference once cropped to the ROI:

1. **Basic augmentation:** Basic augmentation applies random geometric transformations with 50% probability each: random horizontal flip and random vertical flip.
2. **Aggressive augmentation:** In addition to the basic flips, aggressive augmentation applies: random rotation: $\pm 15^\circ$, with border reflection, random zoom/crop: scale factor uniformly sampled from $[0.85, 1.0]$, followed by resize to original dimensions and random shear: horizontal shear factor uniformly sampled from $[-0.1, 0.1]$, applied with 50% probability. This configuration is inspired by the augmentation strategy described in [15], adapted to operate without expanding the dataset size: augmentations are applied during training rather than generating new samples offline.

3.6 Training protocol

All experiments follow the same training protocol to ensure fair comparison:

Dual ViT classifier

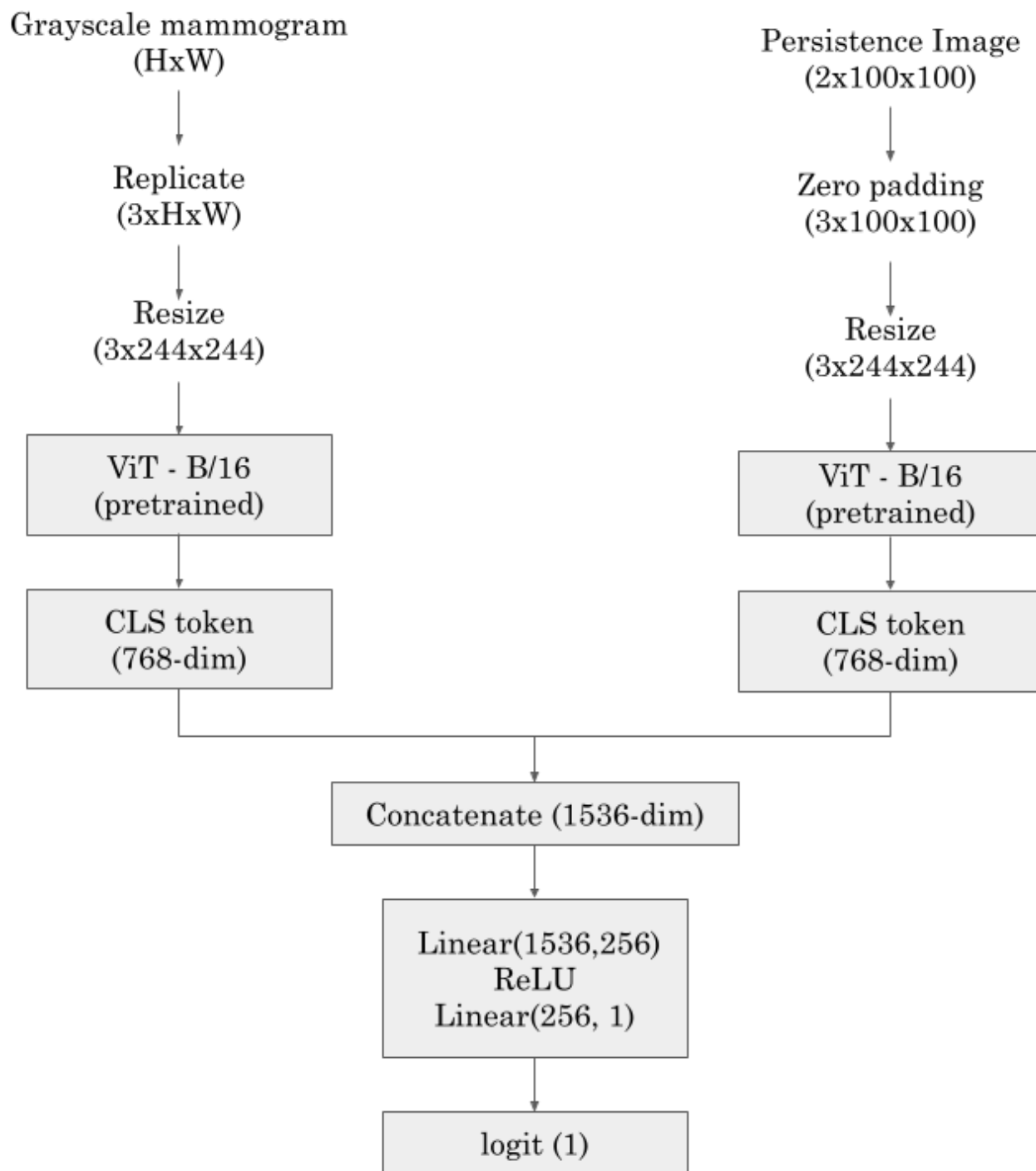


Figure 3.5: Dual-ViT fusion architecture: the mammogram and the persistence image are each processed by a separate pretrained ViT-B/16. The two CLS token embeddings (768-dim each) are concatenated and classified by an MLP.

ViT and TDA vector classifier

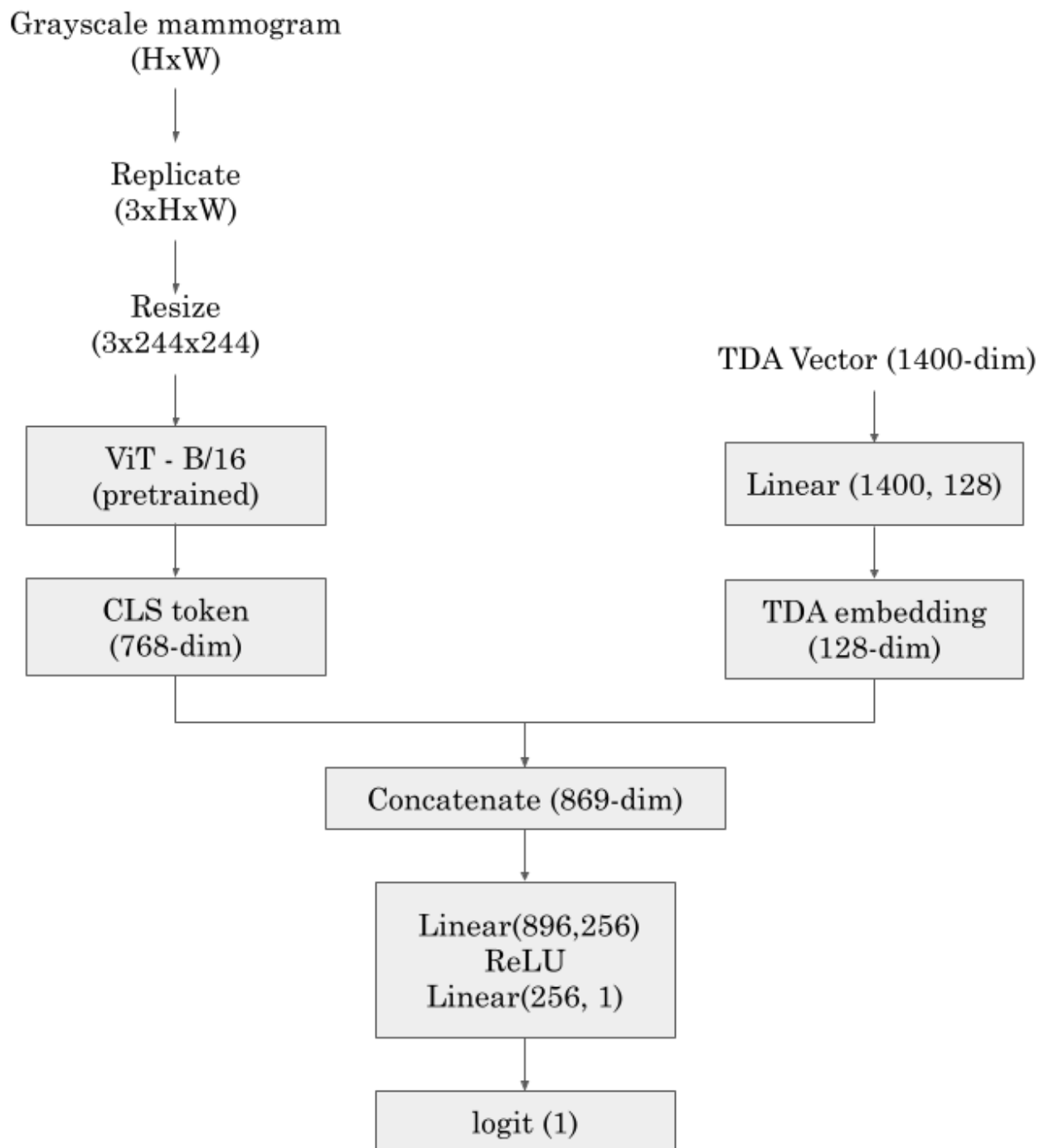


Figure 3.6: Linear fusion architecture: the mammogram is processed by ViT-B/16 (768-dim CLS token), while the 1400-dim TDA vector is projected to 128 dimensions. Both are concatenated and classified by an MLP.

1. **Loss function:** Binary Cross-Entropy with Logits Loss (`BCEWithLogitsLoss`) is used, with a `pos_weight` parameter to handle class imbalance. The weight is computed as the ratio of negative to positive samples in the training set:

$$w_{\text{pos}} = \frac{N_{\text{neg}}}{N_{\text{pos}}} \quad (3.5)$$

2. **Optimizer and learning rate schedule:** the Adam optimizer [39] is used with a learning rate of 10^{-4} . A Cosine Annealing learning rate scheduler [40] is applied over the total number of epochs ($T_{\text{max}} = 30$), gradually reducing the learning rate to zero following a cosine curve.
3. **Early stopping:** to prevent overfitting, early stopping is implemented with a patience of 5 epochs. Training halts if the validation loss does not improve for 5 consecutive epochs. The model checkpoint with the lowest validation loss is retained for final evaluation.
4. **Evaluation metrics:** models are evaluated on the held-out test set using the metrics defined in Section 2.1: Accuracy, F1 Score, and Area Under the ROC Curve (AUC). The AUC is used as the primary metric for comparison, as it is threshold-independent and robust to class imbalance.
5. **Hyperparameters:** table 3.1 summarizes the training hyperparameters, which are shared across all experiments.

Table 3.1: Training hyperparameters

Parameter	Value
Batch size	16
Learning rate	10^{-4}
Optimizer	Adam
Scheduler	Cosine Annealing ($T_{\text{max}} = 30$)
Maximum epochs	30
Early stopping patience	5 epochs
Random seed	2
Classification threshold	0.5

3.7 Experimental design

Seven experiments are designed to systematically evaluate the two hypotheses. They are organized in two groups:

3.7.1 Hypothesis 1: Impact of preprocessing (E1–E3)

- **E1: Baseline ViT:** No preprocessing, basic augmentation (flips only). Uses the ViT classifier (Section 3.4.1). This establishes the baseline performance of a pretrained ViT on raw mammographic crops.
- **E2: ViT with preprocessing:** CLAHE + Gaussian denoise applied, basic augmentation. Same architecture as E1. This isolates the effect of preprocessing.
- **E3: ViT with preprocessing + aggressive augmentation:** Same as E2 but with aggressive augmentation (rotation, zoom, shear). This evaluates whether stronger augmentation can compensate for the small dataset size.

3.7.2 Hypothesis 2: Impact of TDA features (E4–E7)

All TDA experiments use the best configuration from H1 (preprocessing enabled, basic augmentation) as their baseline, and add topological information through different fusion strategies:

- **E4: Dual-ViT with persistence images (cropped lesion):** Uses the DualViTFusionModel (Section 3.4.2). The persistence image is computed from the pre-cropped lesion ROI.
- **E5: Dual-ViT with persistence images (masked mammogram):** Same architecture as E4, but the persistence image is computed from the masked mammogram crop (full mammogram $\times$ ROI mask).
- **E6: Linear fusion with TDA vectors (cropped lesion):** Uses the FusionModel (Section 3.4.3). The 1400-dimensional compact descriptor is computed from the pre-cropped lesion ROI.
- **E7: Linear fusion with TDA vectors (masked mammogram):** Same architecture as E6, but the compact descriptor is computed from the masked mammogram crop.

3.7.3 Threshold optimization

The experiments above evaluate models at the default decision threshold of 0.5. However, this threshold is an arbitrary choice that can be tuned after training and it does not affect the model itself, only how its continuous probability outputs are converted into binary predictions.

This is particularly relevant in clinical contexts: in breast cancer screening, a false negative (missing a malignant case) carries far greater risk than a false positive (an unnecessary biopsy). This is why a last experiment is performed: E8 takes the best-performing model and tunes the decision threshold on the validation set using Youden’s J statistic: it maximizes sensitivity + specificity - 1 [41]. Then, it evaluates on the test set with the selected threshold. The model weights are identical: no retraining is performed, only the decision boundary changes.

Table 3.2 summarizes the experimental configurations.

Table 3.2: Summary of experimental configurations

Experiment	Preprocessing	Augmentation	TDA input	Architecture
E1	No	Basic	—	ViT classifier
E2	Yes	Basic	—	ViT classifier
E3	Yes	Aggressive	—	ViT classifier
E4	Yes	Basic	PI (crop)	Dual-ViT
E5	Yes	Basic	PI (masked)	Dual-ViT
E6	Yes	Basic	Vector (crop)	Linear fusion
E7	Yes	Basic	Vector (masked)	Linear fusion
E8	?	?	?	?

3.8 Implementation and reproducibility

The project is implemented as a modular Python codebase with the following structure:

```

src/
├─ datasets/
│   └─ experiment_dataset.py      # PyTorch Dataset class
├─ models/
│   └─ vit_model.py               # ViT classifier
│   └─ fusion_model.py           # DualViT and Linear fusion
├─ tda/
│   └─ tda_features.py           # TDA extraction functions
├─ training/
│   └─ experiment_runner.py      # Training loop and evaluation
└─ utils/
    └─ image_utils.py            # DICOM loading and path resolution
    └─ preprocess.py             # CLAHE + denoise pipeline

```

All experiments are orchestrated from a Jupyter notebook (`notebook.ipynb`) that calls the modular components. Results (metrics, learning curves, and predictions) are persisted to JSON files in a `checkpoints/` directory, enabling survival across kernel restarts and reproducibility of the analysis.

Looking at it file by file, the structure is:

- **notebook.ipynb:** notebook that describes the project and triggers the experiment execution.
- **src/utils/:**
 - **image_utils.py:** has seven functions to take care of image loading and path resolution:

- * `load_dicom(path)`: given a full path, it uses `pydicom` [32] to load the image and transform it into a `numpy` matrix [42]. It normalizes the image to have values between 0 and 255.
- * `get_all_dcm_files_with_size(rel_path)`: given a relative path, it builds the full path using the `BASE_PATH` defined in the same `image_utils.py` file, whilst taking into account the long path limitations in Windows OS. With the `os` module it extracts all files with `.dcm` extension in said path, and sorts them by size (as described above, it will be used to distinguish masks from cropped mammograms).
- * `resolve_mammogram_path(rel_path)`: extracts the mammogram path from a relative path, taking into account that the file name might be wrong. If more than one files are found in the provided folder, it raises an exception.
- * `resolve_cropped_mammogram_path(rel_path)`: extracts the cropped mammogram path from a relative path, taking into account that the file name might be wrong. If one file is found in the path, it returns the path. If two files are found, it returns the path to the smallest one. If more than two files are found in the provided folder, it raises an exception.
- * `resolve_roi_mask_path(rel_path)`: extracts the ROI mask path from a relative path, taking into account that the file name might be wrong. If one file is found in the path, it returns the path. If two files are found, it returns the path to the biggest one. If more than two files are found in the provided folder, it raises an exception.
- * `cache_cropped_mammogram_images_as_np_arrays(save_dir, df)`: given a dataset with "cropped image file path" and "pathology" columns, it (1) resolves all the paths to the cropped mammogram images (2) converts them into `numpy` arrays and saves them in the `save_dir` provided and (3) saves a `.csv` file in that same folder with two columns: path to the converted image and the corresponding label, that is 1 if it is malignant and 0 if it is either benign or Benign without callback. This function is used to reduce the time loading `dcm` files for every execution.
- * `extract_roi_crop(mammogram, mask, padding_ratio=0.15)`: extracts the ROI crop from a mammogram and applies the mask to it.
- **preprocess.py**: contains three functions for image preprocessing:
 - * `apply_clahe(image)`: applies Contrast Limited Adaptive Histogram Equalization (CLAHE) [19] to a grayscale image using `cv2.createCLAHE` [34] with `clipLimit=2.0` and `tileGridSize=(8,8)` to enhance local contrast.
 - * `denoise(image)`: applies a Gaussian blur with a 5×5 kernel using `cv2.GaussianBlur` [34] to reduce noise.
 - * `preprocess(image)`: chains `apply_clahe` followed by `denoise`, returning the pre-processed image.
- **src/models/**:
 - **vit_model.py**: defines the `ViTClassifier` class, a `PyTorch nn.Module`[31] that wraps a pretrained Vision Transformer. It uses `timm.create_model` [38] to load

- `vit_base_patch16_224` [24] (ViT-B/16, pretrained on ImageNet). The classification head is replaced with a single `nn.Linear` [31] layer that maps from the ViT's hidden dimension (768) to 1 output (binary classification logit). The forward method passes the input through the full ViT and returns the raw logit (during training, `BCEWithLogitsLoss` [31] applies sigmoid internally) 3.4.
- **`fusion_model.py`**: defines two fusion architectures for combining ViT image features with TDA topological features:
 - * **`FusionModel`**: combines a Vision Transformer to process the images with a Linear Layer to process the vectorial topological descriptors. It loads the same `vit_base_patch16_224` backbone but replaces the head with `nn.Identity` [31] to extract the 768-dimensional CLS token. A separate TDA branch (`tda_fc`, an `nn.Linear` layer) projects the flattened TDA vector to 128 dimensions. The forward method concatenates both feature vectors using `torch.cat` [31] and passes the result through a classifier head: 3.6. The constructor accepts a `tda_input_dim` parameter (default 1400) to match the vector descriptor dimensionality.
 - * **`DualViTFusionModel`**: combines a Vision Transformer to process the images with another Vision Transformer to process the Persistence Images (PI). Both of them are separate `vit_base_patch16_224` branches. Both heads are replaced with `nn.Identity` to extract 768-dim CLS tokens. The forward method adapts the PI input (shape batch, 2, 100, 100 for H0 and H1 homology dimensions) to 3-channel ViT input by padding with a zero channel and resizing to 224×224 using `F.interpolate` [31]; further details are provided in Section 3.7. The two CLS tokens are concatenated via `torch.cat` and passed through a classifier: 3.5.
 - **`src/datasets/experiment_dataset.py`**: defines the `ExperimentDataset` class, a PyTorch Dataset that loads precomputed `.npy` images using `numpy.load` [42]. Constructor parameters:
 - `image_paths`, `labels`: lists of image paths and binary labels.
 - `tda_paths` (defaults to `None`): list of paths to precomputed TDA `.npy` files. When provided, `__getitem__` returns a 3-tuple (`image`, `tda`, `label`) instead of (`image`, `label`).
 - `use_preprocessing` (defaults to `False`): if `True`, applies the CLAHE + denoise pipeline from `preprocess.py`.
 - `augment` (defaults to `False`): if `True`, applies basic augmentation: random horizontal flip via `numpy.fliplr` and random vertical flip via `numpy.flipud`.
 - `aggressive_augmentation` (defaults to `False`): if `True`, applies additional augmentation on top of basic flips: random rotation (15) via `cv2.getRotationMatrix2D` and `cv2.warpAffine`, random zoom/crop (85–100% scale), and random shear (0.1).
 - `tda_as_image` (defaults to `False`): if `True`, the loaded TDA array is kept as a multi-dimensional array (for `DualViTFusionModel`). If `False`, it is flattened to a 1D vector (for `FusionModel`).

- `img_size` (defaults to 224): images are resized to `img_size x img_size` using `cv2.resize`, stacked to 3 channels, normalized to `[0, 1]`, and converted to `torch.Tensor`.
- **`src/training/experiment_runner.py`**: contains the full training pipeline with four functions:
 - `get_class_weights(labels)`: computes the weight for each class for BCEWithLogitsLoss to use, as the ratio of negative to positive samples, to address class imbalance in the dataset.
 - `train_one_epoch(model, dataloader, optimizer, criterion, device)`: runs one training epoch. Calls `model.train()`, iterates over batches, computes loss, back-propagates, and updates weights. Handles both 2-tuple (ViT-only) and 3-tuple (fusion) batches.
 - `evaluate(model, dataloader, criterion, device)`: runs evaluation with `model.eval()` and `torch.no_grad`, to prevent computing the gradient, since `Tensor.backward()` is not used. Computes loss, applies `torch.sigmoid` to get probabilities, and calculates accuracy, F1 score, and AUC-ROC using scikit-learn [33]. Returns metrics, true labels, and predicted probabilities.
 - `run_experiment(model_fn, train_dataset, test_dataset, config)`: orchestrates a full experiment. Splits training data into train/val using StratifiedShuffleSplit (85/15). This guarantees that both classes (benign and malignant) are spread evenly across both groups, the original train group has 48.33% malignant cases and the test one has 51.32%, which is slightly imbalanced, but this study uses the default train-test separation for ease of comparison). Creates DataLoaders, initializes the model, loss (with class weights), Adam optimizer) and CosineAnnealingLR scheduler. Trains with early stopping (patience-based on validation loss), saves the best model checkpoint using `torch.save` in the `checkpoints` folder, evaluates on the test set, and persists all results (metrics, training history, predictions) to a JSON file using `json.dump` in the `checkpoints` folder, for survival across kernel restarts.
- **`src/tda/tda_features.py`**: defines six functions for computing topological descriptors from grayscale images:
 - `compute_persistence_image(image)`: uses CubicalPersistence [35] to compute persistent homology (tracking connected components and holes across intensity thresholds). It transforms the resulting persistence diagram into a fixed-size PersistenceImage [35] of shape (2, 100, 100): one 100x100 image per homology dimension (H0 and H1).
 - `compute_vector_descriptors(image)`: uses `gudhi.CubicalComplex` [36] to compute persistent homology, then extracts compact topological summaries for each homology dimension (H0 and H1) using BettiCurve, Landscape (5 landscapes), and Silhouette [37], all at resolution 100. Returns a 1D numpy array of 1400 dimensions (2×100 Betti + 2×500 Landscape + 2×100 Silhouette).
 - `precompute_tda_cropped_image(cache_csv_path, tda_dir)`: iterates over cached `.numpy` crops, computes persistence images via `compute_persistence_image`, saves each as `.numpy`, and adds a `tda_crop_path` column to the cache CSV.

- `precompute_tda_masked_mammography(cache_csv_path, df, tda_dir)`: loads full mammograms and ROI masks from the original DataFrame, applies `extract_roi_crop` to produce masked crops, computes persistence images, saves them, and adds a `tda_masked_path` column to the cache CSV.
- `precompute_tda_vector_descriptors_cropped(cache_csv_path, tda_dir)`: same as `precompute_tda_cropped_image` but uses `compute_vector_descriptors` instead. Adds a `tda_vec_crop_path` column.
- `precompute_tda_vector_descriptors_masked(cache_csv_path, df, tda_dir)`: same as `precompute_tda_masked_mammography` but uses `compute_vector_descriptors` instead. Adds a `tda_vec_masked_path` column.
- **checkpoints**: contains the files persisted by `run_experiment()` after each experiment: a json file with the results, as well as the best checkpoint obtained, in `pt` format. Each json file contains: test metrics (loss, accuracy, F1, AUC), training history (train loss, val loss, val AUC per epoch), the checkpoint path, and the raw test predictions (`y_true`, `y_prob`) for confusion matrices and ROC curves.

The full code can be found in the zip file attached to this document, or in github: https://github.com/EstherRH00/TDA_Visual_Transformer. The `README.md` file contains all setup and running instructions, and the notebook has been uploaded in html format so that the results are easy to access. All experiments were executed on a single machine with GPU acceleration (CUDA). The random seed is fixed at 2 for all stochastic operations (weight initialization, data splitting, augmentation) to ensure reproducibility.

Chapter 4

Statement on the use of generative artificial intelligence

Generative AI was used; in particular, Claude [43] to do four tasks:

1. Review orthography.
2. Transfer the information from notebook markdown to tex format. Then, it was reviewed by the author.
3. Add documentation to the code in form of function and class comments. It was reviewed by the author before committing.
4. Write the file by file analysis: it was given the `image_utils.py` text as example and was asked to write the rest. Then, it was reviewed by the author.

Chapter 5

Results

This chapter presents the quantitative results obtained from the eight experiments described in Section 3.7. Results are reported on the held-out test set using the best model checkpoint selected by early stopping on the validation set.

Table 5.1 summarizes the test set performance of all experiments in terms of Accuracy, F1 Score, and AUC.

Table 5.1: Test set results for all experiments

Exp.	Description	ACC	F_1	AUC	Sens.	Spec.
E1	Baseline ViT (no preprocessing)	0.3995	0.5505	0.5346	0.9456	0.0519
E2	ViT + preprocessing	0.6720	0.6196	0.7228	0.6871	0.6623
E3	ViT + preprocessing + aggressive augmentation	0.6138	0.5101	0.6556	0.5170	0.6753
E4	Dual-ViT + PI (cropped lesion)	0.5820	0.3923	0.5687	0.3469	0.7316
E5	Dual-ViT + PI (masked mammo-gram)	0.5926	0.4500	0.5792	0.4286	0.6970
E6	Linear fusion + TDA vector (cropped)	0.6243	0.4892	0.6337	0.4626	0.7273
E7	Linear fusion + TDA vector (masked)	0.5582	0.5015	0.5787	0.5714	0.5498
E8	E2 + threshold optimization	0.6852	0.5911	0.7228	0.5850	0.7489

The best overall accuracy is achieved by **E8** (ViT with CLAHE + Gaussian denoise, threshold optimization), with an AUC of 0.7228, accuracy of 0.6852, and specificity of 0.7489. However, **E2** achieves the best sensitivity (0.6871) and F1 (0.6196) at the default threshold of 0.5, making it the best configuration from a cancer detection standpoint.

5.1 Hypothesis 1: Effect of preprocessing

Figure 5.1 shows the comparison of the three experiments addressing H1.

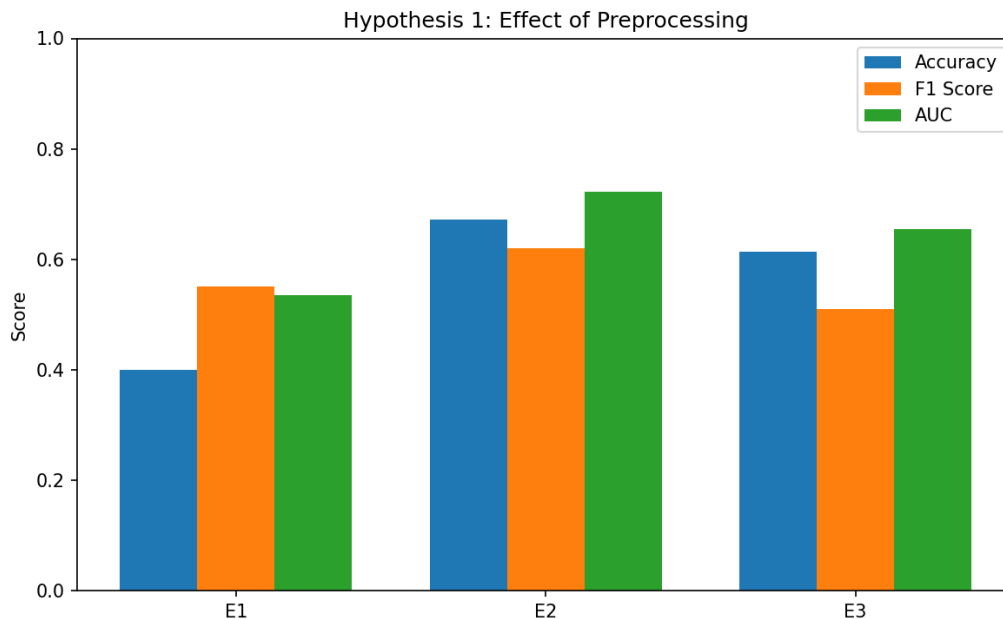


Figure 5.1: AUC, Accuracy, and F1 comparison for experiments E1–E3 (Hypothesis 1).

The introduction of preprocessing (E1 to E2) produces a substantial improvement across all metrics: AUC increases from 0.5346 to 0.7228 (+35.2% relative), accuracy from 0.3995 to 0.6720, and F1 from 0.5505 to 0.6196.

Adding aggressive augmentation (E2 to E3) degrades performance: AUC drops from 0.7228 to 0.6556, accuracy from 0.6720 to 0.6138, and F1 from 0.6196 to 0.5101.

5.2 Hypothesis 2: Effect of TDA features

Figure 5.2 shows the comparison of the four TDA experiments against the E2 baseline.

None of the four TDA-enhanced experiments surpasses the E2 baseline:

- The dual-ViT experiments (E4, E5) achieve AUC values of 0.5687 and 0.5792, respectively, well below the 0.7228 baseline.
- The linear fusion experiments (E6, E7) achieve AUC values of 0.6337 and 0.5787, closer to but still below the baseline.
- Among TDA experiments, E6 (linear fusion with cropped lesion vectors) performs best.

5.3 Threshold optimization

E8 uses the same trained model as E2 (the best-performing configuration) and optimizes the decision threshold on the validation set using Youden’s J statistic [41], which selects the threshold that maximizes Sensitivity + Specificity – 1. Geometrically, this corresponds to the operating point on the ROC curve farthest from the random-guess diagonal.

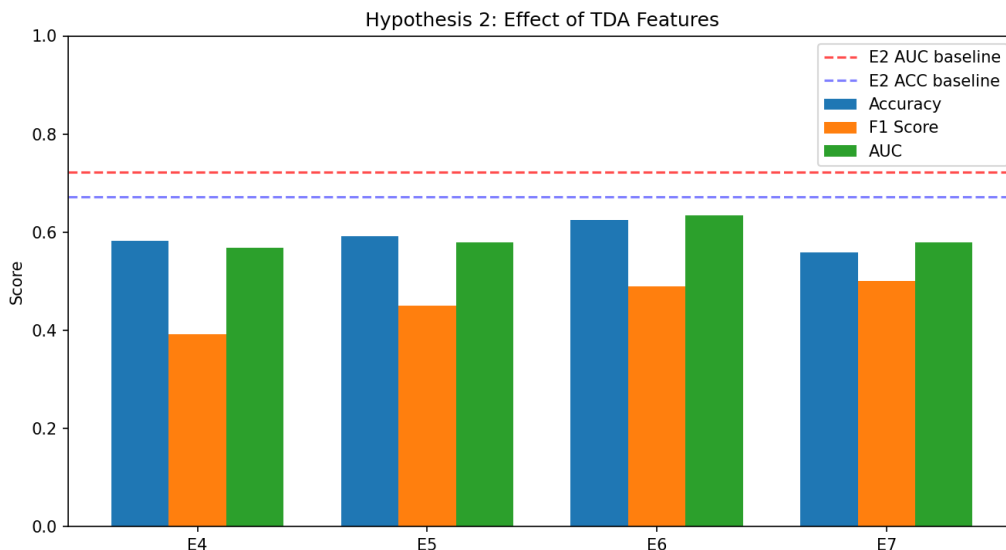


Figure 5.2: AUC, Accuracy, and F1 comparison for experiments E4–E7 (Hypothesis 2) against the E2 baseline (dashed line).

The optimal threshold found on the validation set was 0.59, compared to the default 0.5 used in E1 to E7. Table 5.2 shows the effect of this change on the test set.

Table 5.2: E2 vs. E8: effect of threshold optimization on test set metrics

Config	Threshold	ACC	F_1	AUC	Sens.	Spec.
E2 (default)	0.50	0.6720	0.6196	0.7228	0.6871	0.6623
E8 (Youden’s J)	0.59	0.6852	0.5911	0.7228	0.5850	0.7489

The AUC remains identical (0.7228) since it is threshold-independent and the underlying model is the same. Accuracy improves slightly (0.6720 to 0.6852) and specificity increases substantially (0.6623 to 0.7489), meaning the model correctly identifies $\sim 75\%$ of benign cases. However, sensitivity decreases (0.6871 to 0.5850): the higher threshold requires stronger evidence before predicting malignant, resulting in more missed cancers.

The threshold moved *up* from 0.5 to 0.59, indicating that the model’s raw probabilities were biased toward higher values due to the `pos_weight` parameter used during training. Youden’s J corrects for this by requiring stronger confidence before a malignant prediction.

5.4 Sensitivity and specificity analysis

Table 5.1 includes sensitivity and specificity for all experiments, revealing important patterns in prediction behavior:

- E1 achieves sensitivity of 0.9456 but specificity of only 0.0519: it predicts “malignant” for nearly all samples, making it clinically useless despite a non-zero F1.

- E2 is the only model with balanced sensitivity (0.6871) and specificity (0.6623), confirming it genuinely learned to discriminate between classes.
- E3, E4, E5, and E6 all exhibit high specificity (>0.67) but low sensitivity (<0.52): they are biased toward predicting “benign.” From a clinical screening perspective, this is the most dangerous failure mode, as it results in missed cancers.
- E7 is mediocre in both directions (sensitivity 0.5714, specificity 0.5498).
- E8 further increases specificity (0.7489) at the cost of sensitivity (0.5850), confirming that threshold optimization redistributes errors rather than eliminating them.

5.5 Learning curves

Figure 5.3 shows the training and validation loss curves for all experiments.

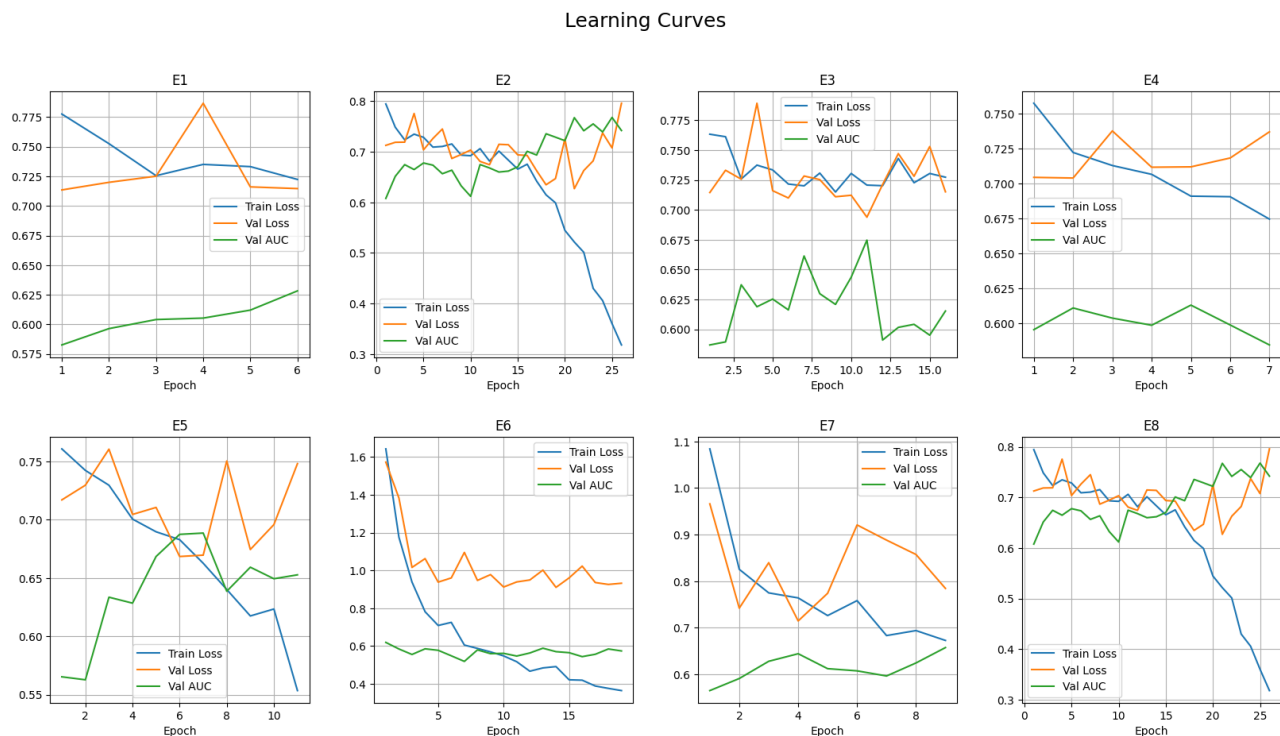


Figure 5.3: Training loss (blue), validation loss (orange) and validation AUC across epochs for all experiments. Early stopping is indicated by the curve termination point.

The learning curves reveal that:

- E1 (no preprocessing) shows minimal learning, with validation loss remaining nearly flat, indicating that the raw images do not provide sufficient signal for the pretrained ViT.
- E2 shows stable convergence with a clear gap between training and validation loss, suggesting moderate overfitting but effective learning.

- E3 (aggressive augmentation) converges more slowly and achieves a higher final validation loss than E2.
- The dual-ViT experiments (E4, E5) exhibit higher variance in validation loss and earlier stopping, consistent with overfitting due to the doubled parameter count.
- The linear fusion experiments (E6, E7) show intermediate behavior between E2 and the dual-ViT models.

5.6 Confusion matrices and ROC curve

Figure 5.4 shows the confusion matrices and figure 5.5 shows ROC curves for all experiments.

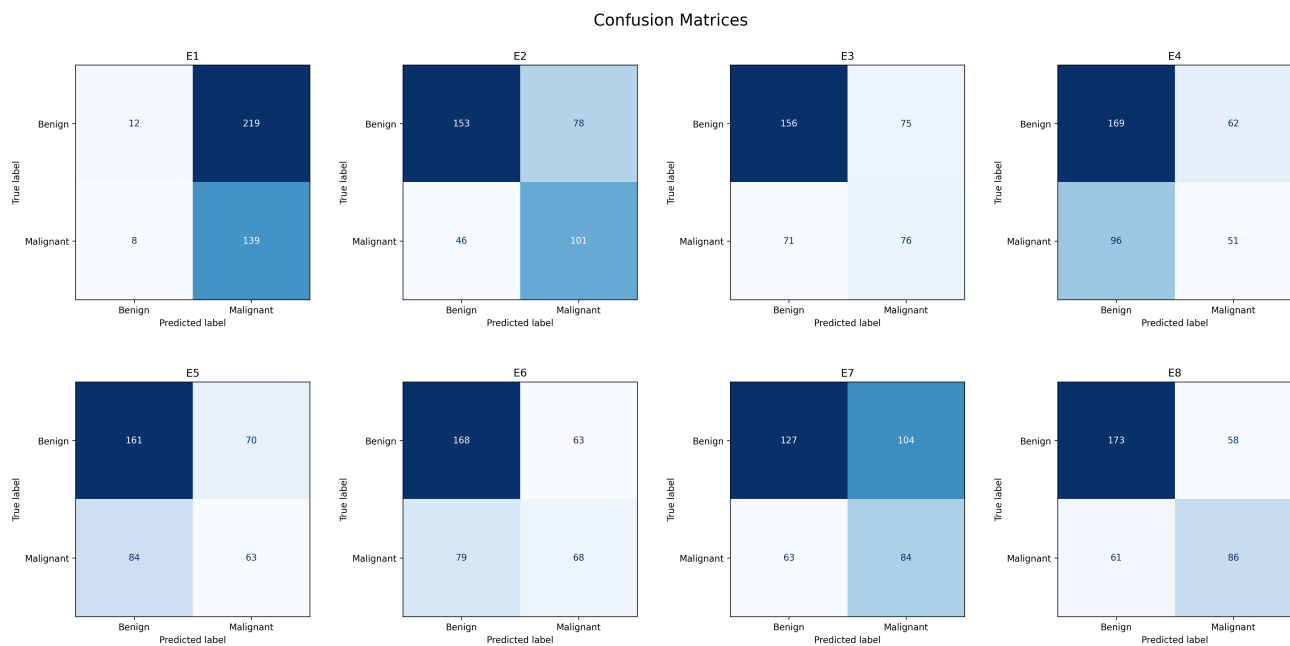


Figure 5.4: Confusion matrices for each experiment

They show that:

- E1 (no preprocessing) is only slightly better than random, and exhibits a strong bias toward 'Malign' predictions.
- E2 exhibits the strongest convergence and misclassifies benign cases as malignant more frequently than the reverse.
- E3 (aggressive augmentation) predominantly predicts 'Benign' and performs slightly better than E1.
- The dual-ViT experiments (E4, E5) are almost identical to E1, but tend to predict 'Benign' instead.

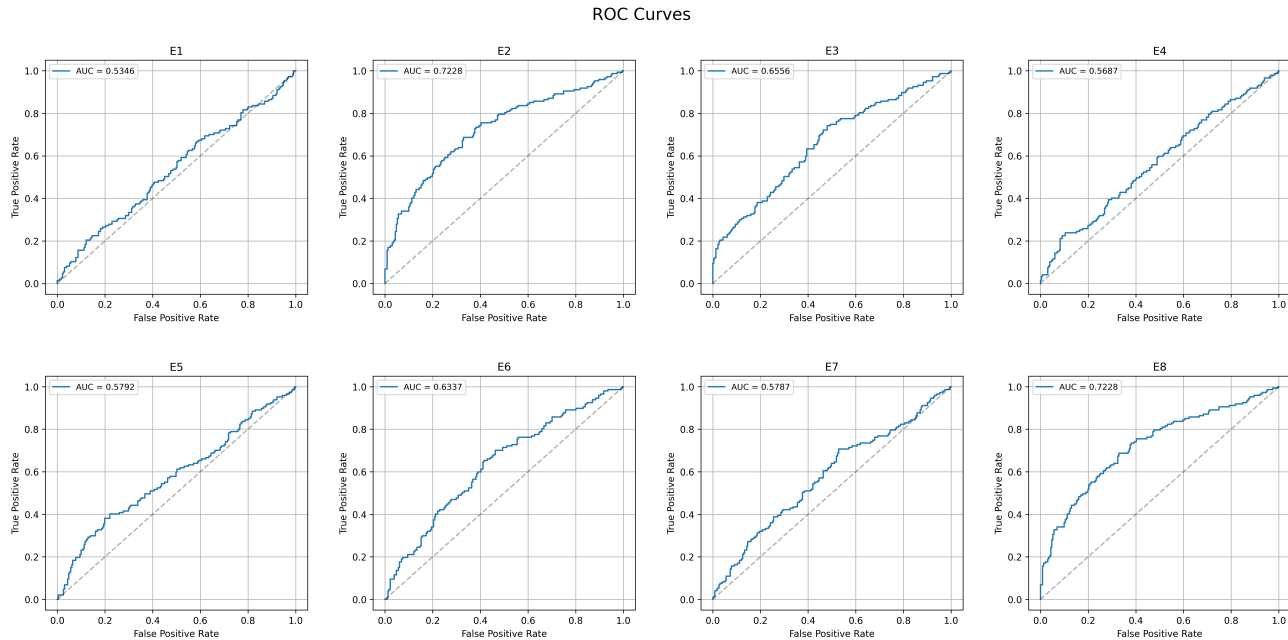


Figure 5.5: ROC curve for each experiment

- The linear fusion experiments (E6, E7) show intermediate behavior between E2 and the dual-ViT models. E6 is almost identical to E5 and E7 predicts in a more *balanced* way but wrongly predicts 'Malign' for benign cases.

5.7 Comparison with the literature

Table 5.3 places the results in context with the studies reviewed in Section 2.1.

Table 5.3: Comparison with CBIS-DDSM binary classification studies from the literature

Study	AUC	ACC	F_1
Transfer Learning and Fine Tuning [15]	0.844	0.844	0.85
EfficientNet-B2 (no augmentation) [21]	0.679	0.6469	-
Deep BI-RADS Network (BiT + attention + text) [22]	0.878	0.796	0.780
This work: E2 (ViT + preprocessing)	0.723	0.672	0.620

The best result of this work (E2, $AUC = 0.723$) is comparable to the EfficientNet-B2 study [21] ($AUC = 0.679$), which also used a small dataset without augmentation. The studies achieving higher performance [15, 22] employed substantially larger augmented datasets (60K images) or additional modalities (textual BI-RADS descriptors).

Chapter 6

Discussion

This chapter interprets the results presented in Chapter 5, analyzes the prediction behavior of the models, and discusses the implications for integrating TDA with Vision Transformers.

6.1 Hypothesis 1: image preprocessing

The introduction of CLAHE + Gaussian denoise in E2 produces the largest performance gain observed in this work: AUC increases from 0.5346 to 0.7228. ViT-B/16 was pretrained on ImageNet, a dataset of high-contrast natural images. Raw mammographic crops have low contrast: intensity differences between lesion and surrounding tissue are subtle. The pretrained patch embedding produces nearly uniform tokens across patches, providing insufficient signal for self-attention. CLAHE redistributes intensity values locally, making structural information accessible to the pretrained features without adding new information.

This is consistent with the literature: [15] also applies CLAHE before pretrained CNNs, and [21], which used only resolution normalization, achieved lower AUC (0.679) despite using EfficientNet-B2.

Regarding aggressive augmentation (E3), the degradation from E2 (AUC 0.7228 to 0.6556) may seem to contradict the success of augmentation in [15]. That study expanded the dataset to 60K images offline before training, whereas E3 applies transformations online to the same 1K samples. On small ROI crops, rotation, zoom and shear distort the diagnostic features that distinguish benign from malignant masses.

6.2 Hypothesis 2: effect of TDA features

None of the four TDA experiments surpass the E2 baseline. Three factors explain this.

- **Redundancy with preprocessed ViT features:** after CLAHE enhances contrast, the ViT's self-attention captures structural relationships such as connectivity, boundary irregularities, texture patterns; that are precisely what TDA also captures through H_0 (connected components) and H_1 (cycles). The topological descriptors become redundant rather than complementary. This is supported by the relative performance: E6 (AUC = 0.6337) improves over E1 (AUC = 0.5346, no preprocessing) but not over E2 (AUC =

0.7228, preprocessing). TDA carries structural information, but it is a subset of what the preprocessed ViT already extracts.

- **Capacity-data mismatch:** with about 1K training samples, additional parameters hurt rather than help: the dual-ViT architecture (E4, E5) doubles parameters to about 172M. The classifier head grows to 393,472 trainable parameters, and the TDA branch must adapt pretrained features to persistence images, that are not similar to the ones in ImageNet. On the other hand, the linear fusion (E6, E7) adds fewer parameters, explaining its better relative performance.
- **Representation gap for persistence images:** persistence images live in birth-death space: position (x, y) represents a feature born at filtration x and dying at y . This has no spatial correspondence to pixel positions. A ViT pretrained on natural images expects spatially coherent input.
- **Cropped lesion vs. masked mammogram:** there is no consistency in performance between topological descriptors computed on cropped images or on masked images. Experiments E4 and E6 used cropped images, while E5 and E7 use masked ones. E4 performed slightly worse than E5, but E6 obtained the best results of all the TDA experiments. This does not give a clear answer to the question raised in the Experiments section 3.7, regarding which of the images would give the best topological descriptors.

The sensitivity/specificity analysis reinforces this interpretation. All TDA models (E4 to E6) exhibit low sensitivity (<0.47) and high specificity (>0.70): they are biased toward predicting benign. This is the clinically most dangerous failure mode in cancer screening. The added TDA parameters appear to make the models more conservative rather than more discriminative, defaulting to the majority-class prediction when uncertain.

The negative result for H2 does not imply that TDA is unsuitable for medical image classification in general. It indicates redundancy in the specific configuration tested: pretrained ViT on preprocessed images with about 1K samples. TDA might provide complementary value in scenarios with larger datasets, without effective preprocessing, with alternative integration strategies (e.g., TDA vectors as additional transformer tokens), or at multiple spatial scales.

6.3 Threshold optimization

E8 demonstrates that post-training threshold optimization can meaningfully shift the operating point of a classifier without retraining. The Youden-optimal threshold (0.59) moved upward from the default 0.5, which is explained by the `pos_weight` parameter used during training: by penalizing false negatives more heavily, the loss function pushes the model's output probabilities upward, making 0.5 an overly permissive threshold that generates excess false positives.

Correcting for this bias via Youden's J improves accuracy (0.6720 to 0.6852) and specificity (0.6623 to 0.7489), but at the cost of sensitivity (0.6871 to 0.5850). This highlights a fundamental tradeoff: threshold optimization cannot improve the model's discriminative ability (AUC remains 0.7228); it can only redistribute errors between false positives and false negatives.

In a clinical screening context, the appropriate threshold depends on the acceptable false negative rate. Youden's J treats both types of errors equally, which may not be appropriate

when missing a cancer carries far greater risk than an unnecessary biopsy. A deployment-oriented approach would select a lower threshold guaranteeing sensitivity ≥ 0.85 , accepting the corresponding reduction in specificity.

6.4 Comparison with the literature

The best result (E2, AUC = 0.723) is competitive with [21] (AUC = 0.679), which also used a small dataset without expansion. Studies achieving higher performance used fundamentally different strategies: 60K augmented images [15] or additional text modalities [22].

This suggests the primary bottleneck is data volume, not model architecture or feature engineering. TDA does not address this bottleneck because it provides a different representation of the same limited data, not additional training signal.

Other limitations that we have found in this project are:

- **Single random seed:** all experiments use seed 2. Results may vary with different initializations.
- **Frozen ViT backbone:** fine-tuning the last transformer layers could improve performance, particularly for TDA experiments.
- **No hyperparameter search:** the same configuration is used for all experiments. Fusion models may benefit from different optimization settings.
- **Breast masses only:** calcifications exhibit different topological characteristics and may yield different results.
- **No statistical significance testing:** without multiple runs, observed differences may be within noise margins.

Chapter 7

Conclusions and future works

This work set out to evaluate whether image preprocessing and topological features improve Vision Transformer classification of mammographic images. The main conclusions are:

1. **H1 is supported.** Preprocessing (CLAHE + denoise) improves AUC from 0.53 to 0.72. This was expected, as pretrained features require adequate input contrast to be effective.
2. **H2 is not supported.** TDA features did not improve over the preprocessed ViT baseline. This was partially unexpected, as the literature suggested topological descriptors could provide complementary structural information, but in practice they proved redundant given the preprocessing and limited dataset size. The sensitivity and specificity analysis further revealed that all TDA models were biased toward predicting benign (sensitivity < 0.47 for E4 to E6), which is the most dangerous failure mode in cancer screening.
3. Aggressive augmentation without dataset expansion hurts performance. This was unexpected given the success of augmentation in the literature, but explained by the difference between offline expansion and online distortion.
4. **Threshold optimization improves accuracy but does not improve discriminative power.** E8 demonstrates that the default threshold of 0.5 was suboptimal for the E2 model due to the effect of `pos_weight` on output probabilities. Youden's J selects a threshold of 0.59 that improves accuracy and specificity but reduces sensitivity. This confirms that the fundamental bottleneck is the model's AUC, not the decision boundary.

Next, we reflect on the objectives defined in Section 1.2.

- To implement a baseline Vision Transformer model for mammographic image classification using the CBIS-DDSM dataset: **achieved** (E1).
- To evaluate the impact of image preprocessing techniques on the performance of the baseline model: **achieved** (E2, E3).
- To extract topological features from mammographic images using Topological Data Analysis methods: **achieved** (persistence images and vector descriptors computed for all samples).

- To design the architecture integrating TDA features with the Vision Transformer model: **achieved** (dual-ViT and linear fusion models).
- To compare all approaches using quantitative performance metrics such as accuracy, sensitivity, specificity, F1-score, and AUC: **achieved** (8 experiments evaluated on ACC, F1, AUC, sensitivity, and specificity).
- To develop a theoretical and practical understanding of Vision Transformer architectures and medical image preprocessing techniques: **achieved** (Chapters 2 and 3).

The schedule defined in Section 1.5 was followed with minor deviations. Because of work constraints, the development phase started a week late but thanks to the margins considered in each task, there was enough time to complete the project. The iterative prototyping methodology was adequate: building from a baseline (E1) and progressively adding components allowed isolating the effect of each modification. No fundamental changes to the methodology were required.

Regarding the impacts identified in Section 1.3:

- **SDG 3 (Health):** the work contributes a reproducible evaluation of ViT-based mammographic classification. While the system is not ready for clinical deployment, it demonstrates that simple preprocessing can make pretrained models viable for this task.
- **SDG 10 (Reduced inequalities):** the pipeline uses a publicly available dataset and open-source tools, making the approach accessible to researchers without proprietary data.
- **Sustainability:** all experiments were executed on a single GPU over approximately 12 hours total. The use of early stopping and pretrained models reduced unnecessary computation.
- **Model bias:** the CBIS-DDSM dataset has limited demographic diversity. The model's performance across different populations remains untested, which is a limitation for any eventual clinical application.

Based on the limitations described in the previous chapter, future lines of work might be:

1. **Dataset expansion:** augment the training set offline to 10K-60K images before training, following the strategy in [15].
2. **ViT backbone fine-tuning:** unfreeze the last N transformer layers to adapt pretrained features to the medical imaging domain.
3. **Alternative TDA integration:** inject TDA vectors as additional tokens in the transformer sequence rather than through a separate branch.
4. **Multi-seed evaluation:** repeat experiments with multiple random seeds to establish statistical significance.
5. **Calcification subset:** extend the study to breast calcifications, which exhibit different topological characteristics.

6. **Explainability analysis:** use attention maps to compare what the ViT attends to with and without preprocessing, and whether TDA regions correlate with attention peaks.

Chapter 8

Glossary

This section defines the most relevant terms and acronyms used in the Report.

Acronyms:

- **ACC**: Accuracy. Proportion of correctly classified instances out of the total.
- **AUC**: Area Under the ROC Curve. Measures a classifier's ability to distinguish between classes across all thresholds.
- **BI-RADS**: Breast Imaging Reporting and Data System. Standardized lexicon for describing mammographic findings.
- **CBIS-DDSM**: Curated Breast Imaging Subset of the Digital Database for Screening Mammography. Public benchmark dataset for breast cancer image analysis.
- **CLAHE**: Contrast Limited Adaptive Histogram Equalization. Local contrast enhancement technique that operates on image tiles with a clipping limit.
- **CLS**: Classification token. Learnable token prepended to the input sequence whose output is used for classification.
- **CNN**: Convolutional Neural Network. Neural network that uses convolutional layers to extract spatial features from images.
- **DICOM**: Digital Imaging and Communications in Medicine. Standard file format for medical images.
- **FN**: False Negative. Positive instance incorrectly classified as negative.
- **FP**: False Positive. Negative instance incorrectly classified as positive.
- **GAN**: Generative Adversarial Network. Architecture where two networks (generator and discriminator) are trained in competition.
- **GELU**: Gaussian Error Linear Unit. Non-linear activation function used in transformer architectures.

- **LLM**: Large Language Model. Neural network trained on large text corpora for language understanding and generation.
- **LN**: Layer Normalization. Normalization technique applied across features within a single sample.
- **MLP**: Multi-Layer Perceptron. Fully connected feed-forward neural network.
- **MRI**: Magnetic Resonance Imaging. Medical imaging technique using magnetic fields and radio waves.
- **MSA**: Multi-Head Self-Attention. Mechanism computing attention between all input tokens using multiple parallel heads.
- **NLP**: Natural Language Processing. Field of AI focused on text and language understanding.
- **PD**: Persistence Diagram. Multiset of (birth, death) points summarizing topological features across a filtration.
- **PI**: Persistence Image. Fixed-size rasterized representation of a persistence diagram.
- **ResNet**: Residual Network. CNN architecture using skip connections to enable training of very deep networks.
- **ROC**: Receiver Operating Characteristic. Curve plotting true positive rate vs. false positive rate across thresholds.
- **ROI**: Region of Interest. Subset of an image containing the relevant structure for analysis.
- **SDG**: Sustainable Development Goal. Set of global objectives defined by the United Nations.
- **TDA**: Topological Data Analysis. Mathematical framework for studying the shape and structure of data.
- **TN**: True Negative. Negative instance correctly classified as negative.
- **TNR**: True Negative Rate. Synonym of specificity.
- **TP**: True Positive. Positive instance correctly classified as positive.
- **TPR**: True Positive Rate. Synonym of sensitivity/recall.
- **ViT**: Vision Transformer. Transformer architecture adapted for image classification by treating image patches as tokens.

Terms:

- **Betti curve**: Discretized function counting the number of topological features alive at each filtration value.

- **Binary classification:** Task of assigning inputs to one of two categories.
- **Cubical complex:** Topological structure built from a grid of pixels, where filtration is defined by intensity values.
- **Decision threshold:** Value (default 0.5) used to convert a model's continuous probability output into a binary prediction. Samples above the threshold are classified as positive.
- **Early stopping:** Training strategy that halts optimization when validation performance stops improving.
- **Filtration:** Sequence of nested topological spaces parameterized by a threshold value.
- **Fine-tuning:** Adapting a pretrained model to a specific task by continuing training on domain-specific data.
- **Homology:** Algebraic structure capturing topological features: H_0 counts connected components, H_1 counts cycles.
- **Landscape:** Piecewise-linear function derived from a persistence diagram by extracting tent functions above the diagonal.
- **Patch embedding:** Linear projection mapping each image patch into a fixed-dimensional token for the transformer.
- **Persistent homology:** Method for tracking topological features as a filtration parameter varies.
- **Pos_weight:** Parameter in BCEWithLogitsLoss assigning higher penalty to false negatives of the minority class.
- **Self-attention:** Mechanism computing pairwise relationships between all tokens in a sequence.
- **Silhouette:** Weighted average of tent functions derived from a persistence diagram.
- **Transfer learning:** Reusing a model trained on one task as the starting point for a different task.
- **Youden's J statistic:** Index for selecting the optimal decision threshold, defined as Sensitivity + Specificity - 1. Maximizes the distance from the random-guess diagonal on the ROC curve.

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